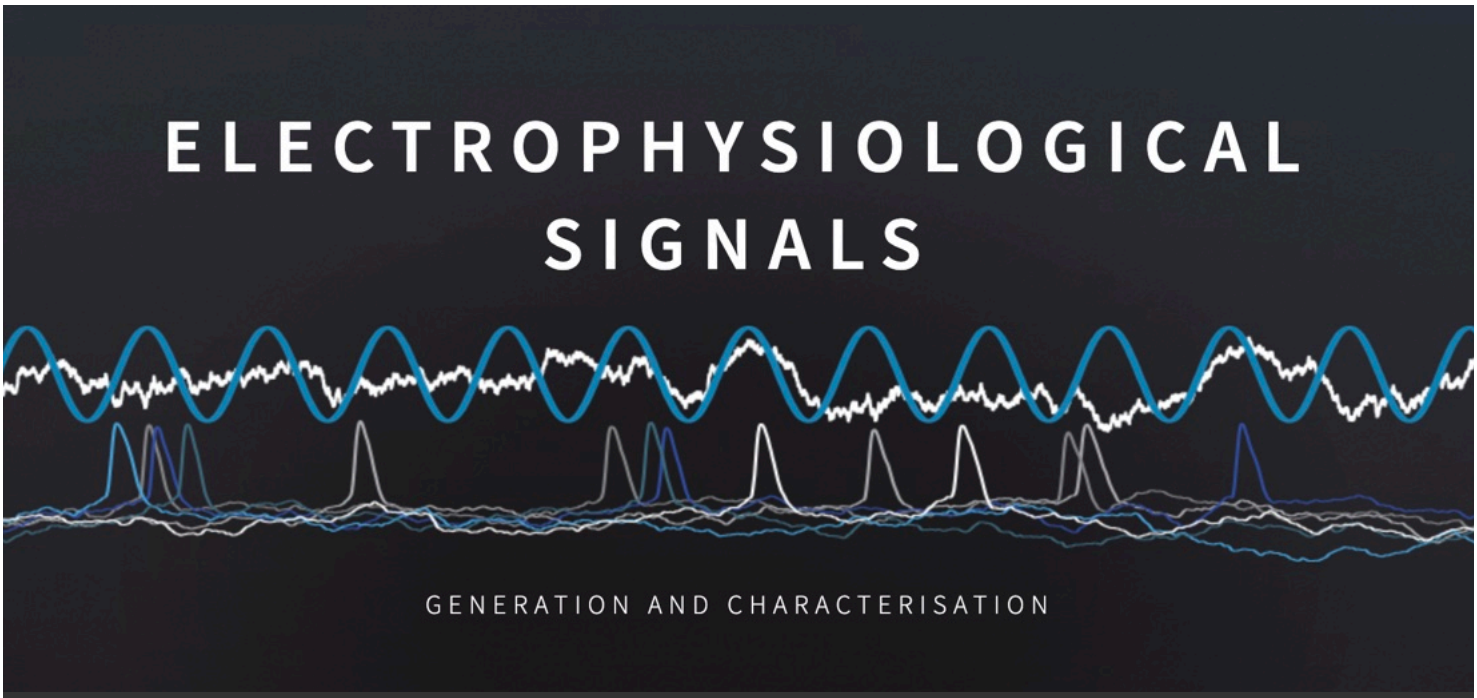


Prof. Ing. Michele GIU**G**LIANO, PhD

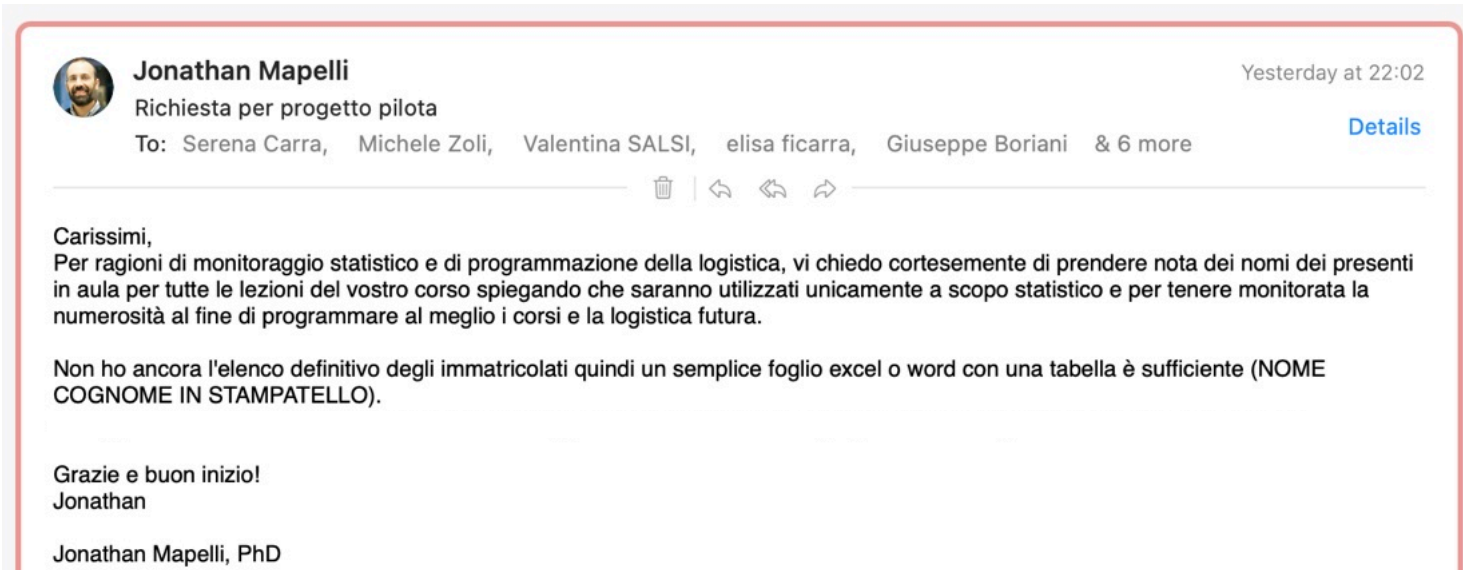
Organization, Logistics, Introduction

CLASS **ATTENDANCE IS NOT** COMPULSORY
BUT **ATTENDANCE TRACKING IS**
please download the app

Download for iPhone

Download for Android

https://www.unimore.it/it/servizi/unimore-app



Jonathan Mapelli Yesterday at 22:02

Richiesta per progetto pilota

To: Serena Carra, Michele Zoli, Valentina SALSI, elisa ficarra, Giuseppe Boriani & 6 more Details

Carissimi,
Per ragioni di monitoraggio statistico e di programmazione della logistica, vi chiedo cortesemente di prendere nota dei nomi dei presenti in aula per tutte le lezioni del vostro corso spiegando che saranno utilizzati unicamente a scopo statistico e per tenere monitorata la numerosità al fine di programmare al meglio i corsi e la logistica futura.

Non ho ancora l'elenco definitivo degli immatricolati quindi un semplice foglio excel o word con una tabella è sufficiente (NOME COGNOME IN STAMPATELLO).

Grazie e buon inizio!
Jonathan

Jonathan Mapelli, PhD

ATTENDANCE TRACKING: **today's code is ???**
(compulsory)

Logistics and organisation

- On Tue: **room M1.6 (35 seats) is free in 8:15-11:00**
- On Wed: **room M1.6 (35 seats) is free in 8:15-9:00**

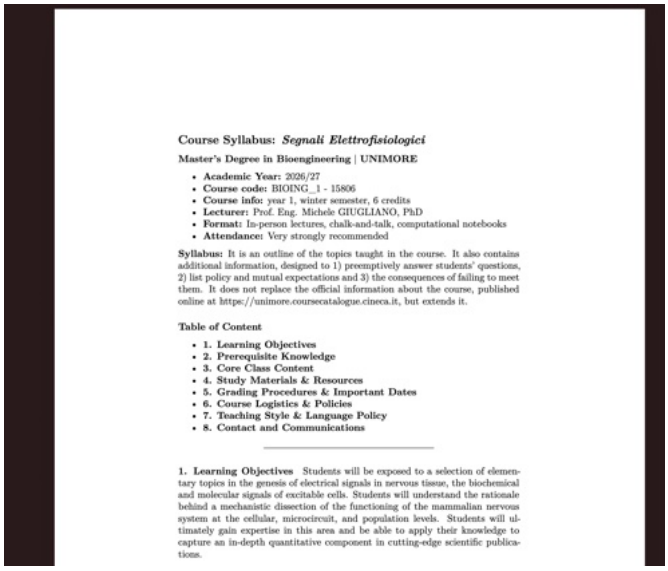
I therefore propose you to

- 1) start classes always at 8:15am and
- 2) move to M1.6 on Tuesdays.
- **We would end our classes on 25 Nov, instead 16 Dec!**

What is a Syllabus?

github.com/mgiu**g**liano/ePhysSignals —> **/overheads**

- **Course Outline:** A brief overview of topics covered in class.
- **FAQs:** Preemptive answers to common doubts/logistics.
- **Mutual Expectations:** Rresponsibilities / commitments required.
- **Accountability & Consequences:** Transparent policies outlining the outcomes of failing them.
- **Clarity:** Upfront, written agreement, to establish shared rules.
- **Recommendations**: Advice, study suggestions, and guidelines rooted in academic honesty.
- **Disclaimer:** Does not replace the official university course catalogue. It complements it.



Course Syllabus: *Segnali Elettrofisiologici*

Master's Degree in Bioengineering | UNIMORE

- **Academic Year:** 2026/27
- **Course code:** BIOING_1 - 15806
- **Course info:** year 1, winter semester, 6 credits
- **Lecturer:** Prof. Eng. Michele GIUGLIANO, PhD
- **Format:** In-person lectures, chalk-and-talk, computational notebooks
- **Attendance:** Very strongly recommended

Syllabus: It is an outline of the topics taught in the course. It also contains additional information, designed to 1) preemptively answer students' questions, 2) list policy and mutual expectations and 3) the consequences of failing to meet them. It does not replace the official information about the course, published online at https://unimore.coursecatalogue.cineca.it, but extends it.

Table of Content

- 1. Learning Objectives
- 2. Prerequisite Knowledge
- 3. Core Class Content
- 4. Study Materials & Resources
- 5. Grading Procedures & Important Dates
- 6. Course Logistics & Policies
- 7. Teaching Style & Language Policy
- 8. Contact and Communications

1. Learning Objectives Students will be exposed to a selection of elementary topics in the genesis of electrical signals in nervous tissue, the biochemical and molecular signals of excitable cells. Students will understand the rationale behind a mechanistic dissection of the functioning of the mammalian nervous system at the cellular, microcircuit, and population levels. Students will ultimately gain expertise in this area and be able to apply their knowledge to capture an in-depth quantitative component in cutting-edge scientific publications.

Tenere questo corso in Inglese?

- **Inclusività**: in presenza anche di un singolo studente internazionale, il corso si tiene comunque in Inglese, a garanzia di un pieno accesso inclusivo alla conoscenza che è libera e globale;
- **Realismo:** l'aula è un microcosmo professionale globale, dove l'Inglese è lo standard operativo;
- **Apprendimento passivo:** assimilazione naturale di terminologia e concetti chiave, a lezione;
- **Requisito strategico:** *skill* indispensabile per industria di alto livello e ricerca (inter)nazionale;
- **Vantaggio:** strumento per colmare il divario tra formazione locale e mercato del lavoro globale;
- **Investimento:** opportunità di sviluppo competenze essenziali per il successo in aziende hi-tech;
- **Basso rischio:** l'aula è un ambiente *protetto* per praticare l'Inglese (attivo o passivo), ben prima di entrare nel mondo "vero" della produzione o della ricerca e sviluppo.

Italiano o Inglese?

- **Flessibilità:** italiano o inglese. La scelta è vostra, ma comporta alcune implicazioni;
- **Reciprocità:** Le partnership Erasmus+ si basano su accordi di scambio internazionali bilaterali (a cui il Docente ha contribuito). La reciprocità è un dovere: UNIMORE deve offrire corsi/traineeship in Inglese. ***L'interesse reciproco di tali accordi viene altrimenti meno.***
- **Patrocinio:** *La preferenza del corso in Inglese* ***facilita notevolmente*** *la capacità del Docente di patrocinare e raccomandare con fiducia gli studenti per eventuali tirocini internazionali, all'interno della* ***propria rete di contatti*** *e ricerca globale.*
- *Se la preferenza del corso è per l'Italiano, gli studenti potrebbero dover fare maggiore affidamento sulle* ***reti degli altri membri del corpo docente*** *di UNIMORE per i collocamenti internazionali.*

POLLME TIME!

Teams - Instant messaging

- Compulsory install
- On your device!
- Enable notifications!
- Check it daily for new posts!
- Check your @unimore email daily!

- THE official way to communicate
- Announcements
- Posts, Links
- Forum / Q&A for the benefit of all

INSTALL & CHECK IT DAILY

Website and GitHub repository

giu**g**liano.info/segnali

https://github.com/mgiu**g**liano/ePhysSignals

DEMO OF THE WEBSITE

Part A: mathematical functions

- Functions and their graph
- Adding a constant or multiplying by a constant
- Notable functions: straight lines, exponentials, and logarithms

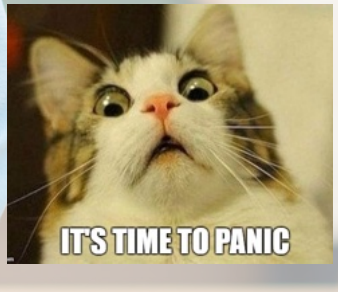
Part B: Derivatives and Integrals

- Intuition for derivatives and elementary functions
- Indefinite and definite integrals

Part C: Mathematical Preliminaries

- The Dirac's Delta-function
- The Convolution Integral
- Trains of Deltas and the Sampling of a function
- Algebraic versus differential equations
- The inhomogenous case
- Numerical solutions
- The Euler's forward method
- Taylor's expansion and comparison with the analytical solution

Prerequisites



Prerequisites

Fundamentals of Circuits and Linear systems

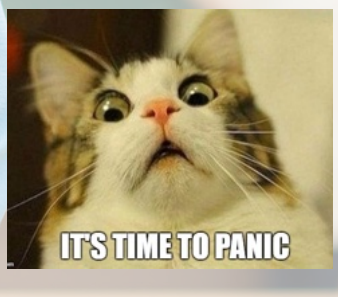
- concept of linearity, concept of zero-state response and forced response
- Fourier and Laplace transforms
- Kirchoff laws, passive components, “solving” circuits, deriving input-output relationships

Elements of Probability Theory

- concept of probability, of expected value, mean, variance
- Bayes’ theorem, random variables

Fundamentals of Scientific Programming

- concept of variable, string, integers, floating points, boolean
- concept of logical operators (AND, OR, NOT)
- for loop, while loop



Familiarity with “Software Carpentry” is a plus

- command line interface and unix shell
- version control with git, GitHub.com
- text editors, IDEs
- *agentic* coding, co-pilot coding, LLM coding

UNIMORE - https://www.youtube.com/playlist?list=PLaYqeCWLtaIPASI9buRVM3ZXdcpLUyHQ8
UNIPD - https://learn.eduopen.org/eduopenv2/course_details.php?courseid=109
EDUOPEN
https://learn.eduopen.org/eduopenv2/course_details.php?courseid=353
https://learn.eduopen.org/eduopenv2/course_details.php?courseid=64
https://learn.eduopen.org/eduopenv2/course_details.php?courseid=229

MIT
The missing semester - https://missing.csail.mit.edu/

Khan Academy
Algebra - https://www.khanacademy.org/math/algebra-home

Calculus - https://www.khanacademy.org/math/calculus-1
https://www.khanacademy.org/math/calculus-2

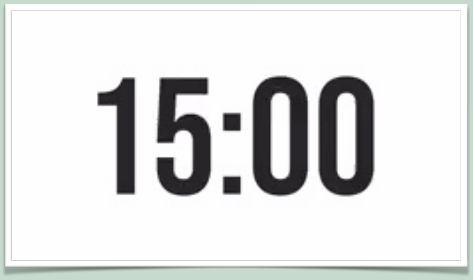
Physics - https://www.khanacademy.org/science/highschool-physics
https://www.khanacademy.org/science/ap-college-physics-1
https://www.khanacademy.org/science/ap-college-physics-2

Electrical Circuits - https://www.khanacademy.org/science/electrical-engineering/ee-circuit-analysis-topic
Computer Science - https://www.khanacademy.org/computing/intro-to-python-fundamentals
Biology - https://www.khanacademy.org/science/biology

Approach me **early**, if you lack fundamental knowledge.

"Diagnostic" questionnaire

Anonymous
15 minutes
Honest self-assessment



https://forms.gle/ysdvSomd8zszipW19

Universitas (*Studium Generale*)

- a small-scale version of the *universe* where **the pursuit of *truth* is a collective act**.
- not a"credential factory"; not a venue for "exam-and-exit"
- "community of scholars" = students as **active participants in expanding human understanding**, not passive vessels waiting for the "correct" answers to pass a test.
- not "study for everyone," but "study of everything": become a ***homo universalis***, someone whose **maturity and breadth of thought** allows them to **serve society in any capacity**, rather than a specialist who only knows how to follow a rubric.
- **"thinking" is an end goal in itself**: by maturing intellectually, you aren't just becoming "employable"; you become a capable steward of culture and society.

Universitas (*Studium Generale*)

- Use AI wisely
- A *multiplication table* (tabelline) forgotten in adulthood leaves a scaffolding. One never learned leaves a void.

Organisation and Practicalities

- Classes ***in person*** - Slides + chalk/board
- **Interactive!** I can't read your mind (yet)
- I do **NOT** mind *stupid* questions (really!)
- **Ask** for explanations, guidance, books, papers, code, hints, intuition, etc.
  - "office hours" are for you! (on appointment)

About “in person” classes

- **Instant feedback:** real-time resolution of complex technical doubts;
- **Social dimension:** enhanced cognitive alignment through non-verbal cues and though joint/shared attention.
- **Mentorship:** personal pathways to projects, theses, professional references.
- **Spontaneous debate:** critical thinking triggered by a *live* academic discourse.
- **Networking:** access to my international network & Erasmus opportunities.
- **Presence:** High-fidelity learning experience, free from digital mediation.

Organisation and Practicalities

- **hand-outs:** available upfront (website/Github)
- **hands-on:** just a web browser (Colab) or (for top students), install ju**py**ter on your device
- **prerequisites:** math, physics, biology, coding

Preparing for the hands-on

- https://youtu.be/GeQAZt4iEgQ (What is Colab?)

OR

- https://github.com/jupyterlab/jupyterlab-desktop
- or a local python/jupyter install on your laptop...
  Ask me or *Claude Code et al.* for help, if you run into trouble!

Exam (for my module) and Evaluation

- **this** module: **written + in-class evals (+ oral)**
- **cum laude** means 32/30 (for this module and overall)
- Overall mark = average **weighted** by CFUs of modules
- Overall mark = **rounded** to closest integer (~~*ceil, floor*~~)
- Overall mark = **requires** all modules (none skipped)
- Each module mark expires after **24 months**

Statistics from the “past”

- Average score of “my” module: **28/30**
- Average score for the “entire” set = **26.5/30**

Exam sessions: 6/year

- Jan 2027, 12th *primo appello invernale*
- Feb 2027, 9th *secondo appello invernale*
- Jun 2027, 8th *primo appello estivo*
- Jul 2027, 15th *secondo appello estivo*
- Sep 2027, 9th *primo appello autunnale*
- Sep 2027, 23rd *secondo appello autunnale*

*Students can register starting **from 15 days before and up until 3 days before** the scheduled exam date.*
*Students who register and **do not show up**, **must skip one session** (i.e., they are not allowed to register to the very next exam session).*

“Full” score registration

necessary conditions

- every other Lecturer does register your score (of their part) officially
- every other Lecturer communicate it “internally” with me

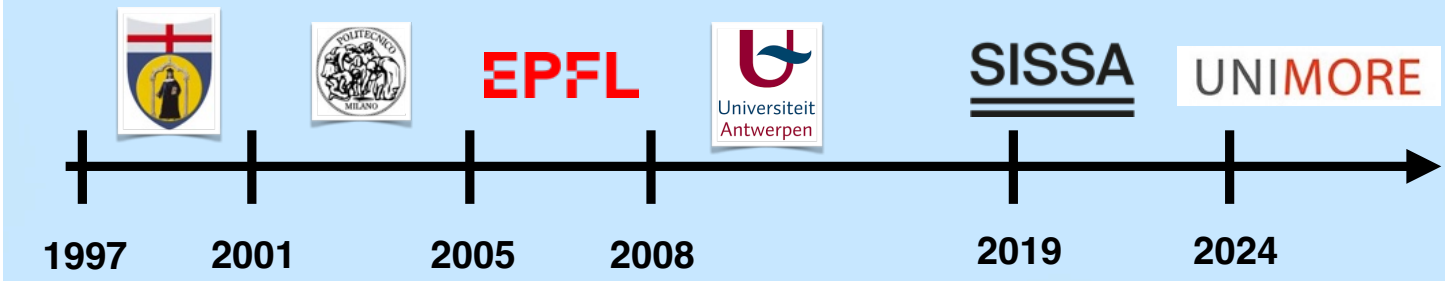
Once per month I check all students if the above qualifying conditions are satisfied and only then I register the “full” score officially by creating an *ad hoc* virtual extra “registration event/session”.

Do not contact me to have the “full” score registered.

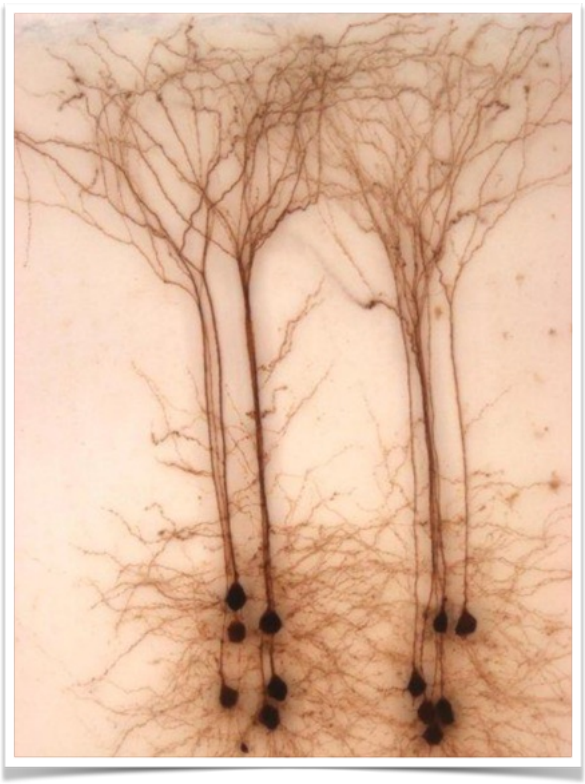
Class organization

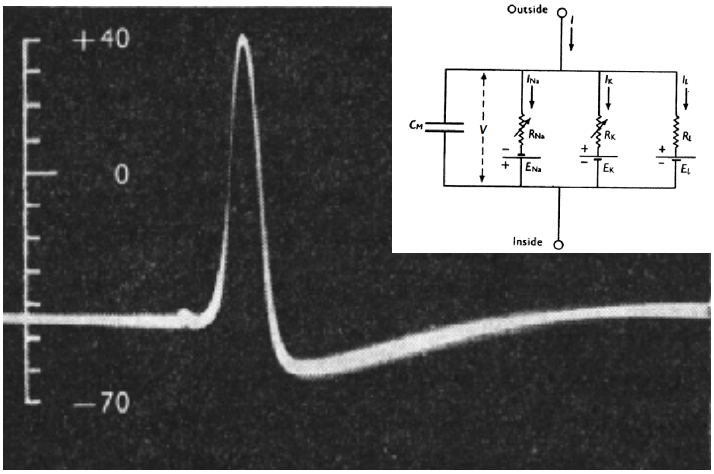
- **Start** *sharp*, on the hour (no academic 15 min)
- **Breaks**: 10 min, every 50 min of lecture

“10-min breaks” means you come back to class in time



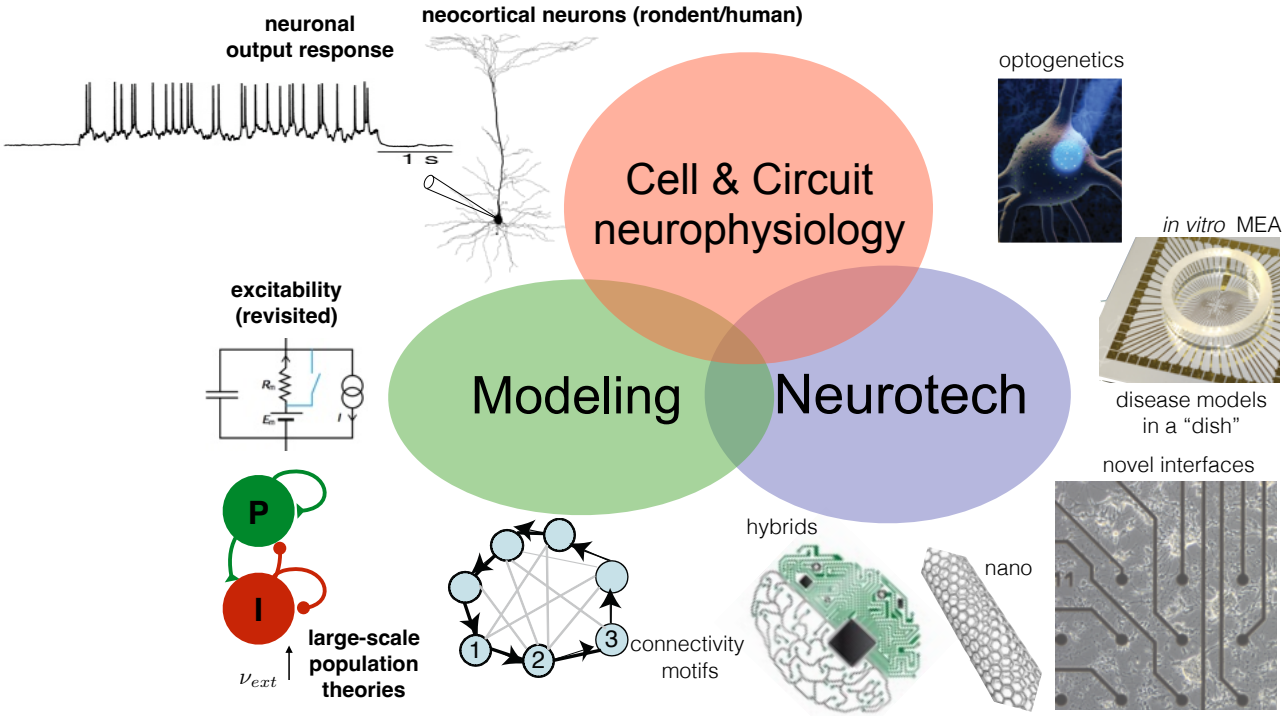
1992-1997:	5-yrs Laurea in **Electronic/Biomedical Engineering**	(Genova)
1997-2001:	PhD in **Bioengineering**	(Milano)
2001-2005:	Postdoc (HSPO) in **Cellular Neurophysiology**	(Bern)
2005-2008:	Group Leader in **Microcircuit Neurophysiology**	(Lausanne)
2008-2019:	Associate and Full Professor of **Neuroscience**	(Antwerpen)
2019- :	Associate Professor of **Physiology**	(Trieste)
2024- :	Associate Professor of **Bioengineering**	(Modena)



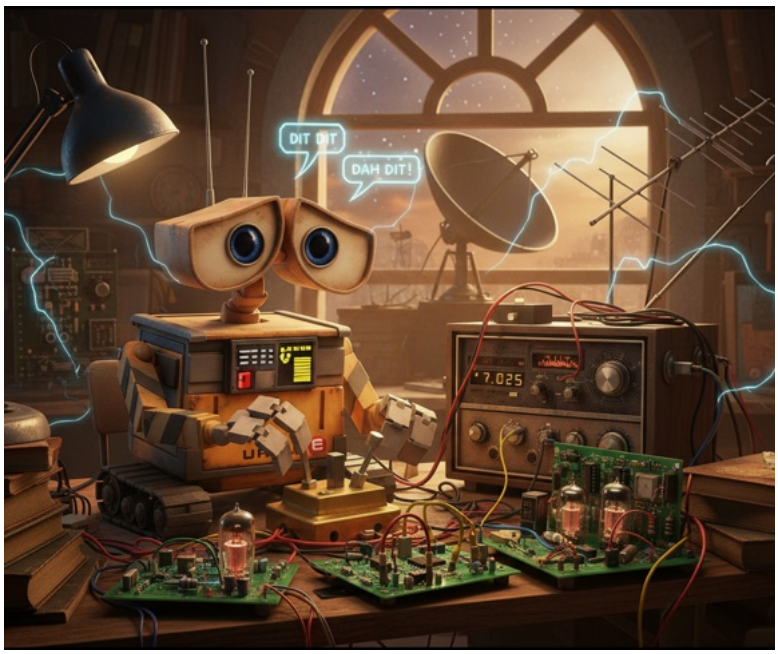


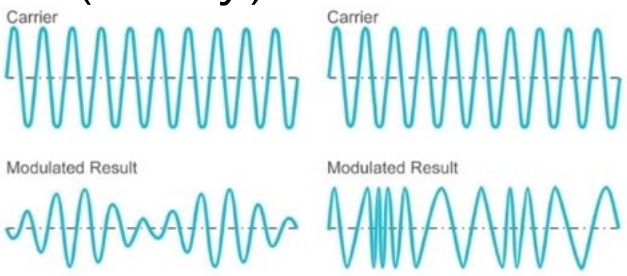
$$\frac{dV}{dt} = -\frac{1}{C_M}\{\bar{g}_K n^4 (V - V_K) + \bar{g}_{Na} m^3 h (V - V_{Na}) + \bar{g}_l (V - V_l)\}$$

Laboratory of Neuroengineering (visits possible)

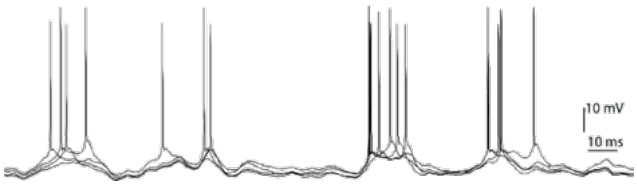


In my "*free*" time (if any)





AM, **FM**



POLL TIME!

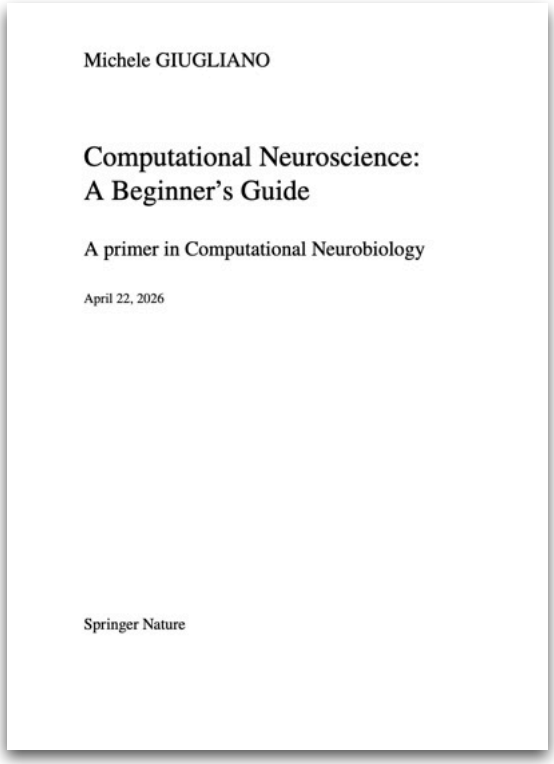
POLL TIME!

Study Material: chapters from

- Halnes et al., (2024) Electric Brain Signals - Cambridge Univ. Press

- Weiss TF (1996) Cellular Biophysics 1-2, MIT Press (ch. 1-5)
- Bear MF, Connors B, Paradiso MA (2006) Neuroscience [..] (ch. 1-5)
- Kandel et al. (2012) Principles of Neural Science, 5th ed., McGraw Hill
- Alberts et al. (2009) Essential Cell Biology, 3rd ed., Garland Science

- Primer on "Brain Facts" (see parts 1-2, 4)
- (few) pages from Longstaff, 2007 (neurons, diversity, CNS)
- "Eye, Brain, and Vision" book by D. Hubel (ch. 1-5)



?

Reading material and Data sample

- *Readings* posted on our website (see *resources*)
- ECG/EMG signals posted on GitHub (see *data*) - follow the tutorial on YouTube and play with the data!

(BIOLOGICAL) SIGNALS

- A "***signal***" is physical variable (or observable), subject to **variations**.
- Variations may be (e.g.) through time **t**, through space **x,y,z**,….
- It is represented as a scalar/vector *function* of **indep.** variable(s).
- Examples of (known) functions: f(**t**) = $sin(2\pi\ 0.1\ \mathbf{t})$ g(**x**,**y**) = $e^{(\mathbf{x}+\mathbf{y})}$
- …a collection of data points, one/more for each value of the indep. var(s).

- *scalar*: EMG (1d), mono music (1d), b/w photo (2d), b/w video (3d), MRI (3d), fMRI (4d)
- *vector*: velocity of a fly (4d —> 3d), stereo music (1d —> 2d), color of a RGB photo (2d —> 3d)

“Animal Electricity” or …conventional Electrical phenomena??

Luigi GALVANI
(1737-1798)

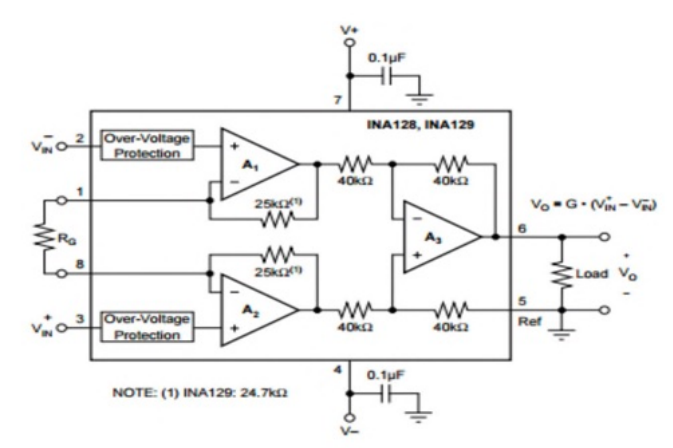
Alessandro VOLTA
(1745-1827)

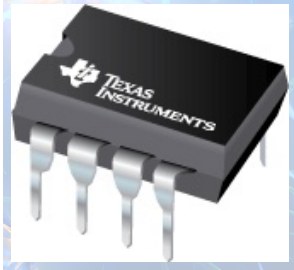
- *connecting* **nerves** to muscles leads to spasms…
- contacting **different metals** together creates an electric field…
- passing a **current** through frog legs, they contract!

Am I… like a frog?

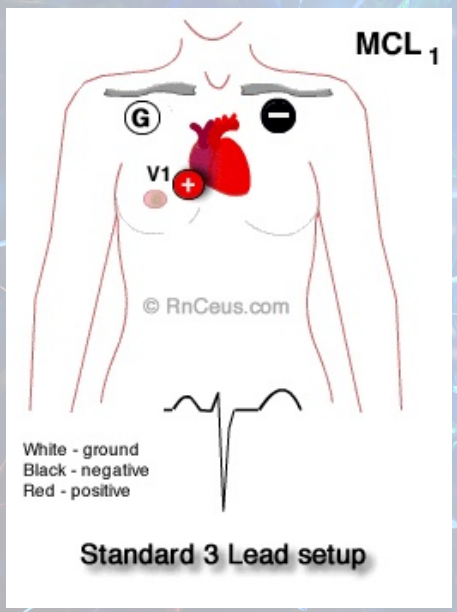
- Why would you need an electronic amplifier?
- Why better a *differential* amplifier?
- What is a *ground-loop?*

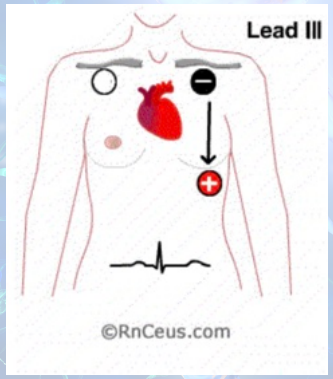
DEMO time!!!

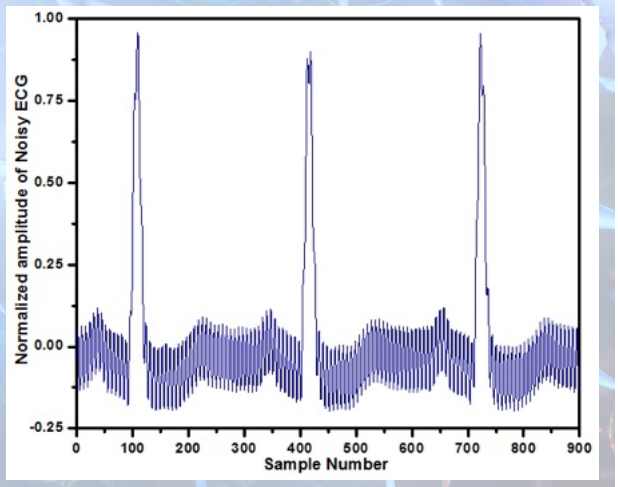




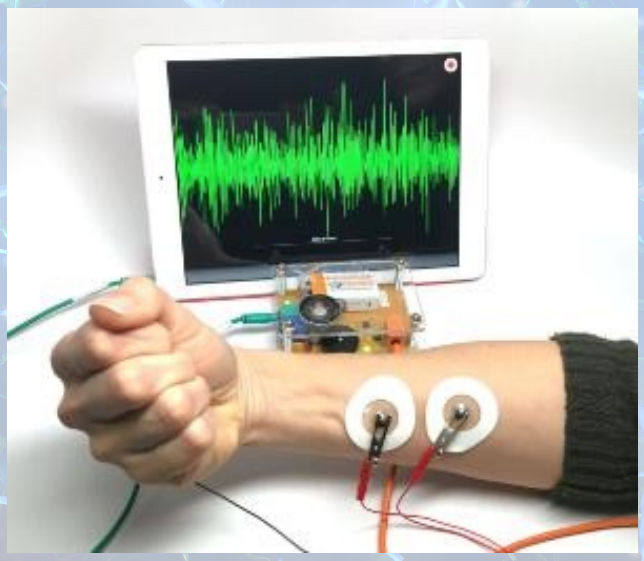
EKG - noise - signals polarity



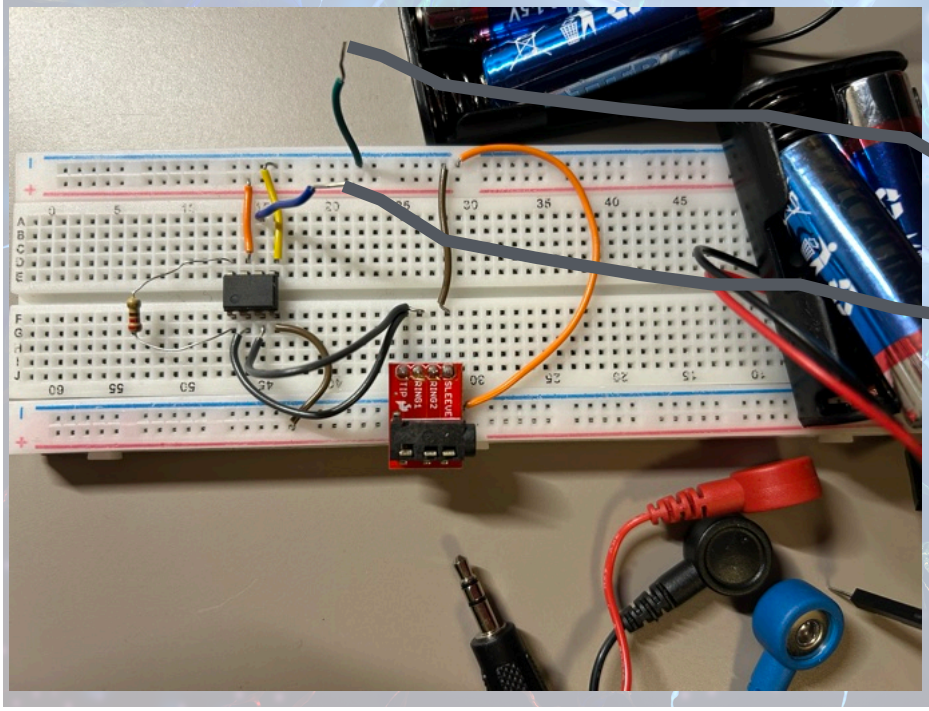




Motor units - (surface) **E**lectro**M**yo**G**raphy



Am I… like a frog?



Play with my recordings! - YouTube Tutorial

- Can you plot the **EKG** and **EMG** traces? Change colors, line style, line thickness? Axis labels?
- Can you infer, from data, the sampling rate (in kHz) in both cases?
- What is the average voltage recorded for each trace? Is it zero-mean? Why? Why not?
- What is the standard deviation of each trace? What is the std dev across time, in 1s-chunks?
- Are those stationary signals? What is the power spectrum of those signals?
- Is there a clear corrupting noise? Why is it at 50 cycle/s? Can you filter it out?
- Can you infer, from data, the Analog-to-Digital conversion resolution (in bits and in mV)?
- What was my heart rate in the EKG measurements? What was the average peak amplitudes?
- What was the variability in my heart rate?
- Does it look like a EKG? Is it upside down? Why? Can you measure R-R, P-R, QT interval, S-T?

BIOLOGICAL SIGNALS: why bother??!

Basic research

Fundamental understanding of the brain, heart, pancreas, muscles…

Understanding (and treating!) dysfunctions

Epilepsy, schizophrenia, Parkinson's, Motor neuron diseases, Multiple Sclerosis, Alzheimer, Blindness, etc. …diabetes, arhythma, etc.

Neuroprostheses, Neuromodulators, …Electroceutics

Retinal/Cochlear implants, (DBS) neuromodulation, BMI, pacemakers, etc.

Reverse engineering the brain

Novel computing paradigms, robotics, computer vision, ML, NeuroAI,….

Brain (dys)functions

Men ought to know that from nothing else but the brain comes joys, delights, laughter and sports, and sorrows, and griefs, despondency, and lamentations.

*And by this, in a especial manner, we **acquire wisdom and knowledge,** and **see** and **hear** and **know** what are foul and what are fair, what are bad and what are good, what are **sweet** and what are **unsavory**...*

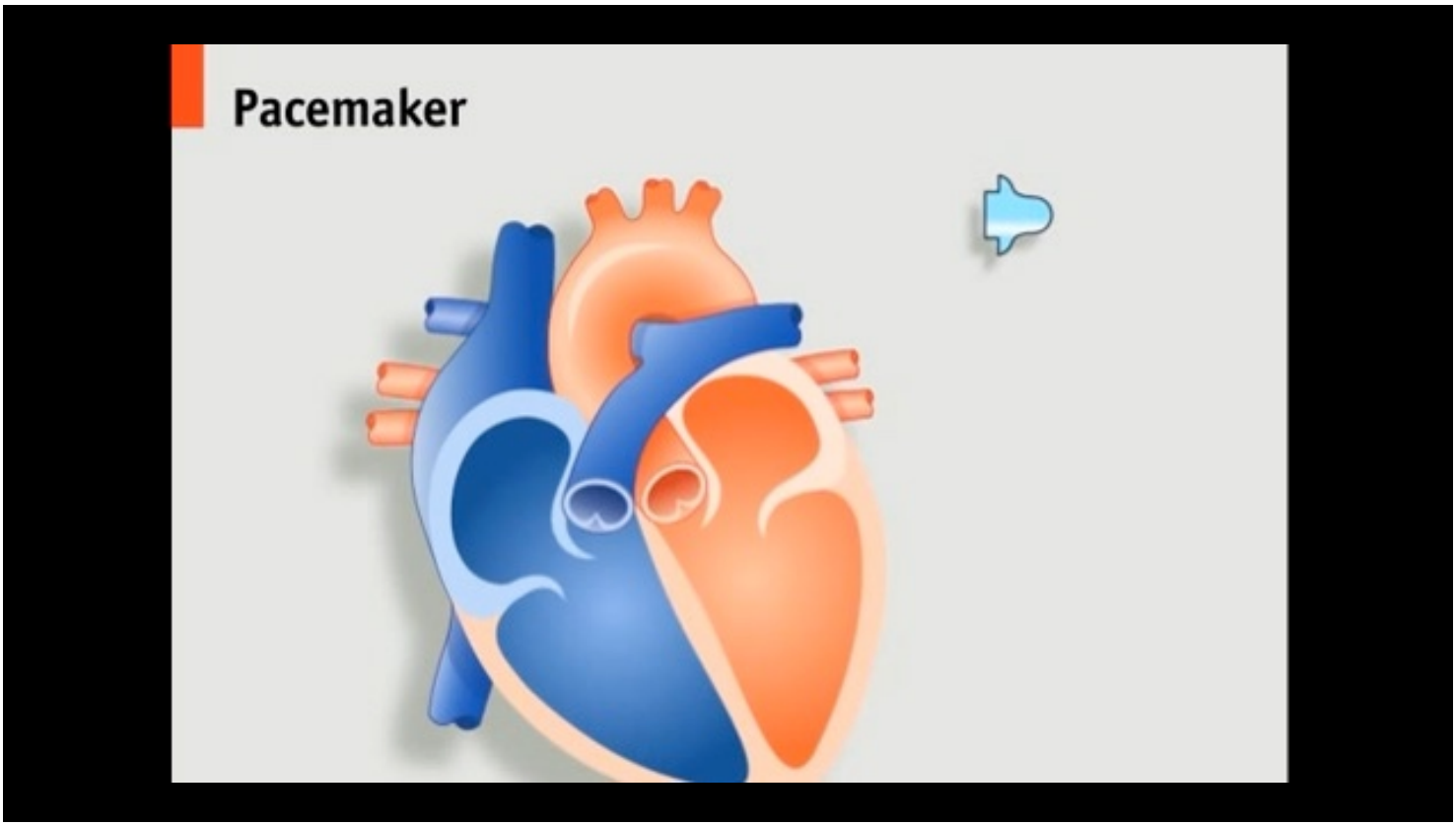
*And by the same organ we become **mad** and **delirious**, and **fears** and **terrors** assail us...*

*All these things we endure from the brain when it is **not healthy**... In these ways I am of the opinion that **the brain exercises the greatest power in man**.*

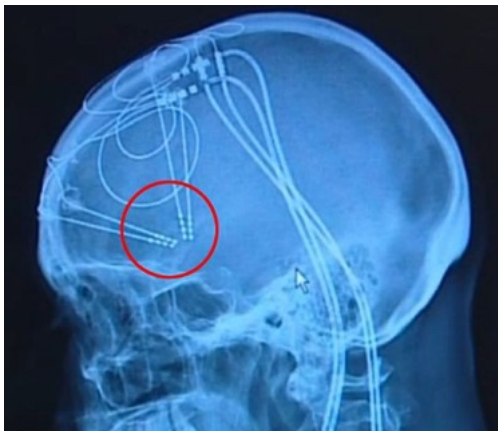
Hippocrates (fourth century B.C.): "*On the Sacred disease*"

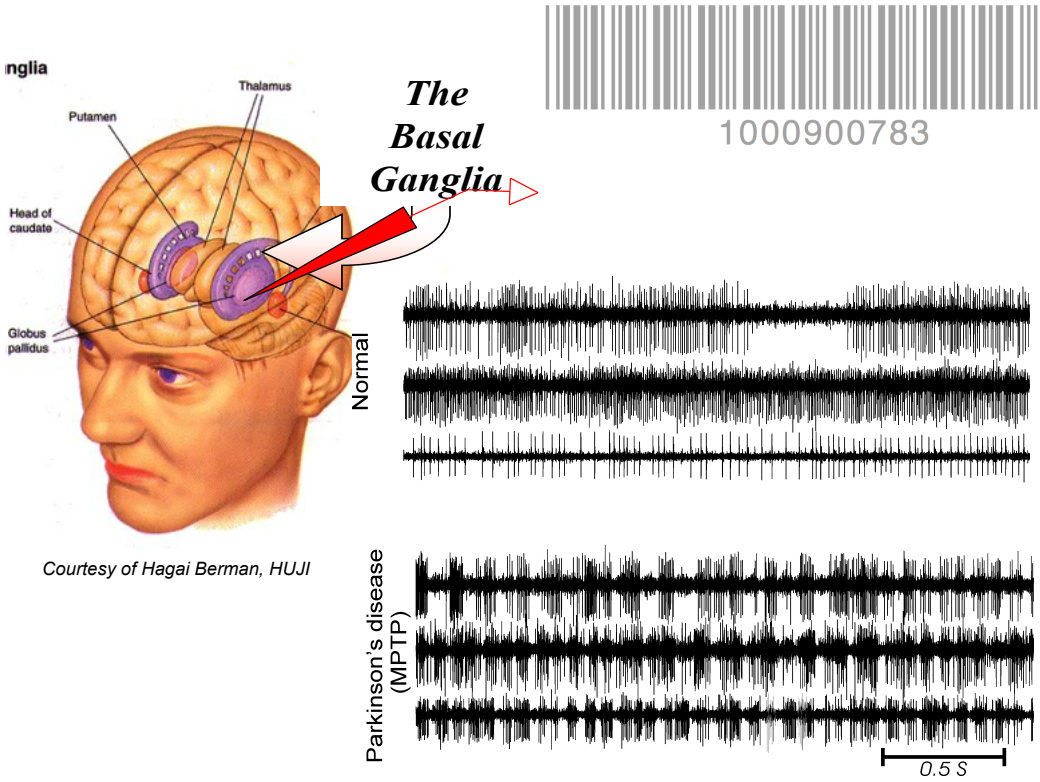
Anyone you know with...

- a heart pacemaker? an implantable defibrillator?
- a cochlear implant?
- a deep-brain stimulator?



Beyond Neuropharmaca: Neuromodulation & Electroceutics





1000900783

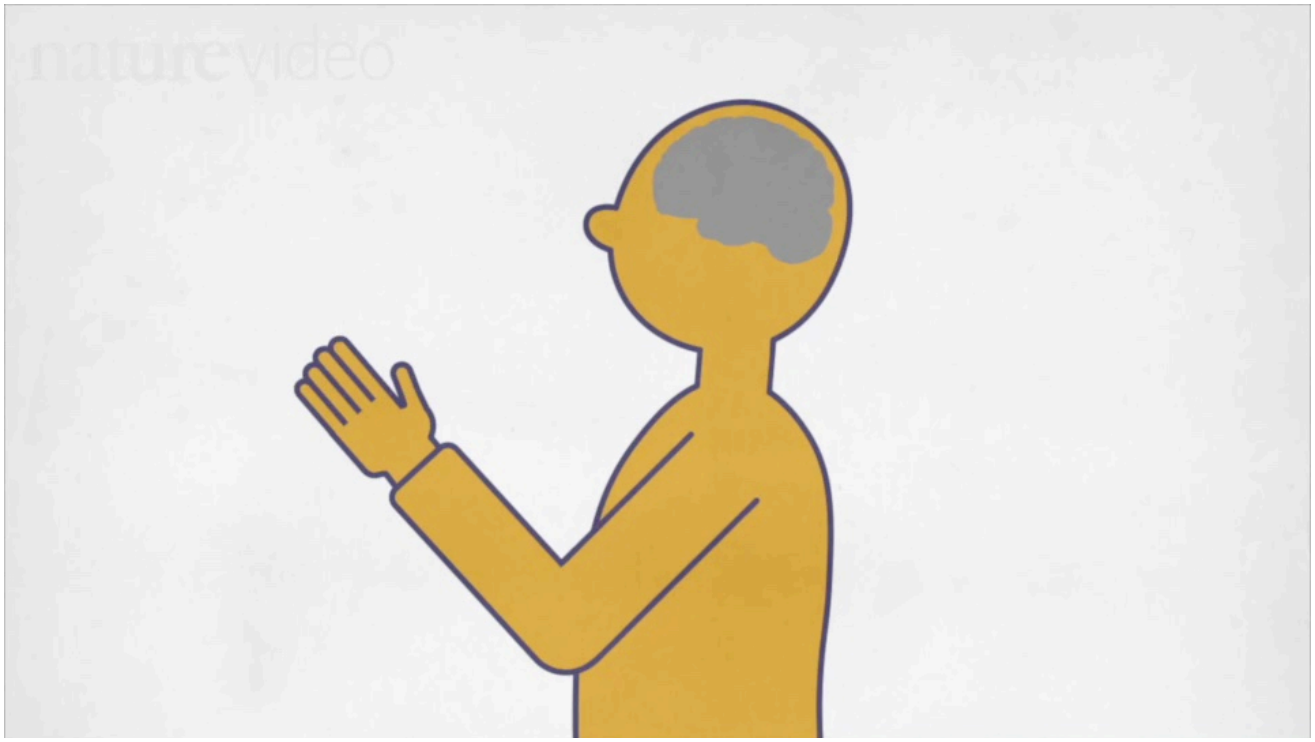
Courtesy of Hagai Berman, HUJI

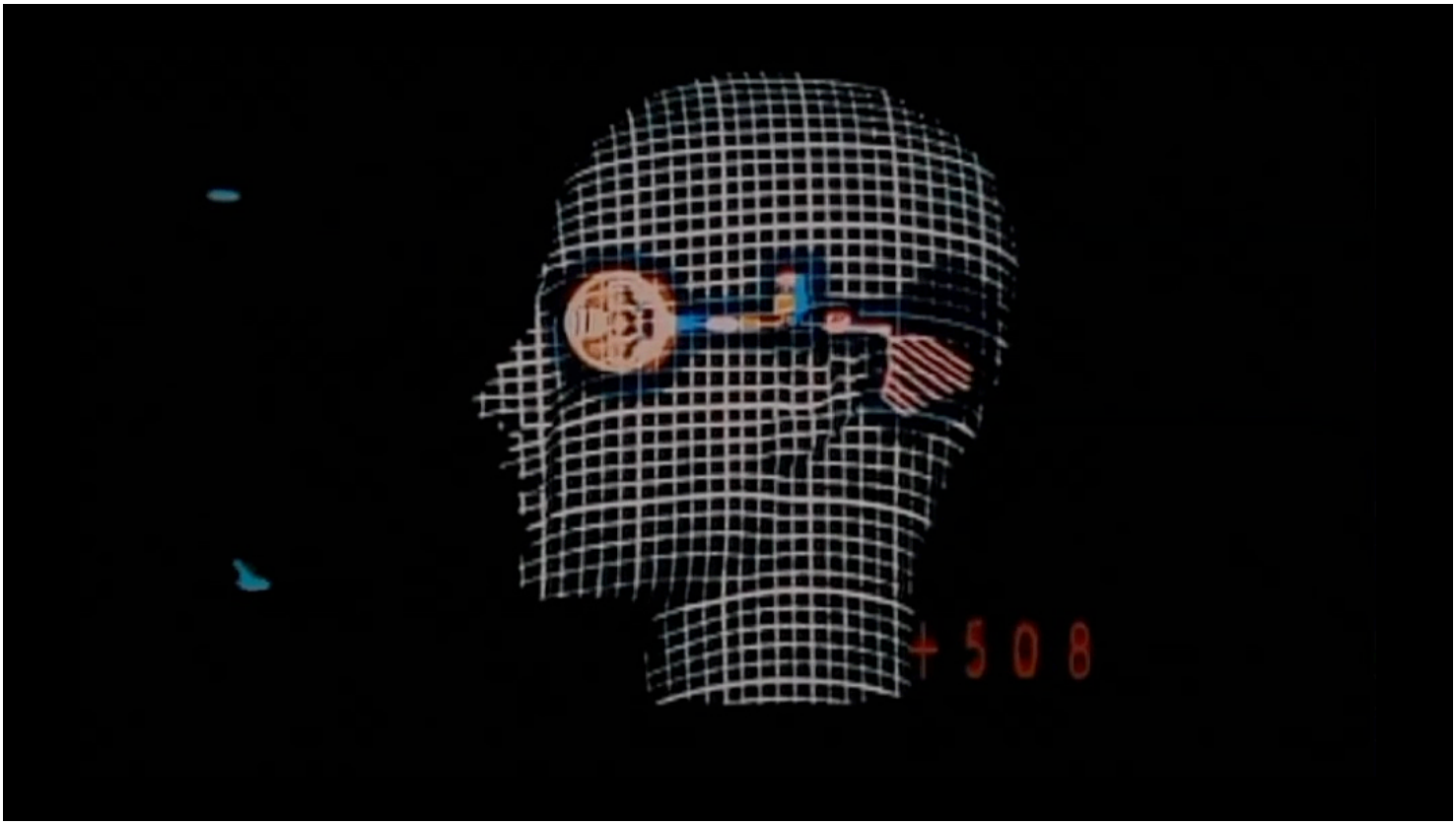
(high-frequency, electrical) Deep Brain (extracellular) Stimulation

MEDTRONIC's Activa DBS Therapy

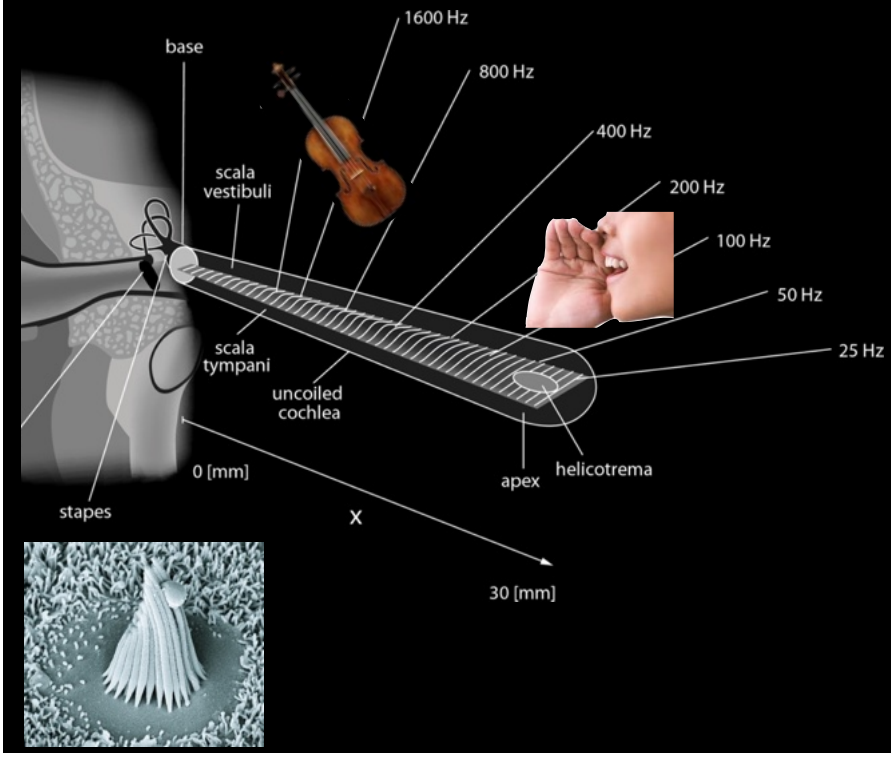
Bionics and Neuroprosthetics

Motor (& Sensory) Neuroprosthetics



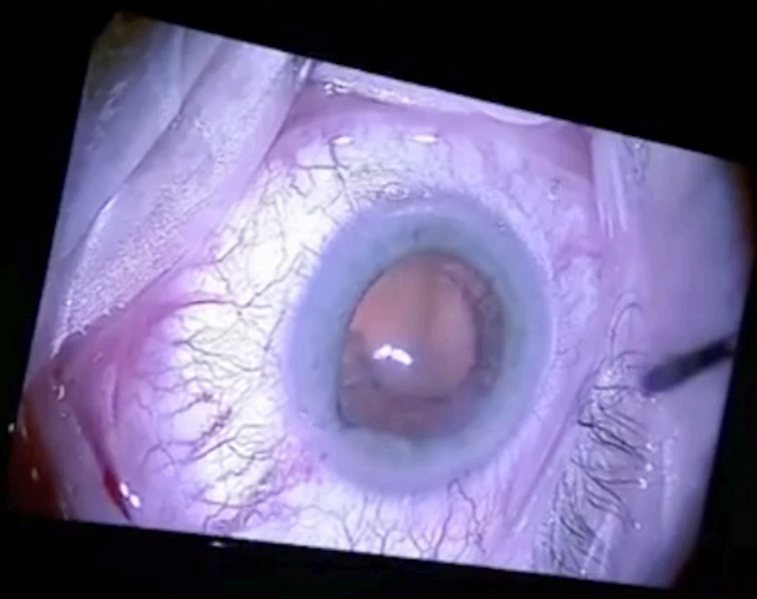


Sensory Neuroprosthetics



Cochlear

Sensory Neuroprosthetics

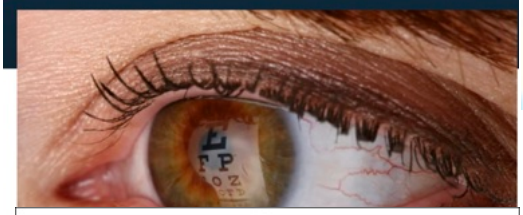


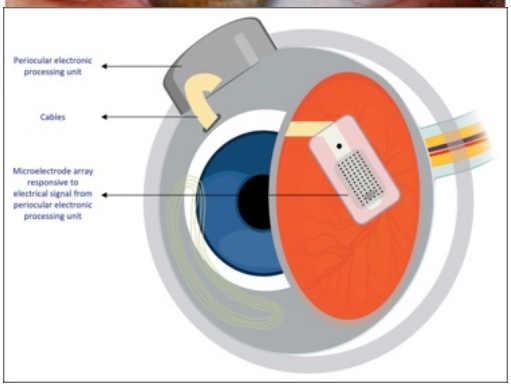
MEDTECH

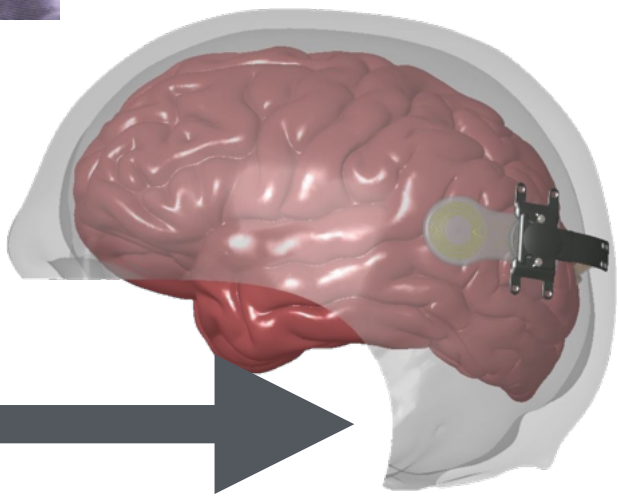
'Fantastic technology and a lousy company': Second Sight's 'bionic eyes' have gone obsolete while still implanted, report finds

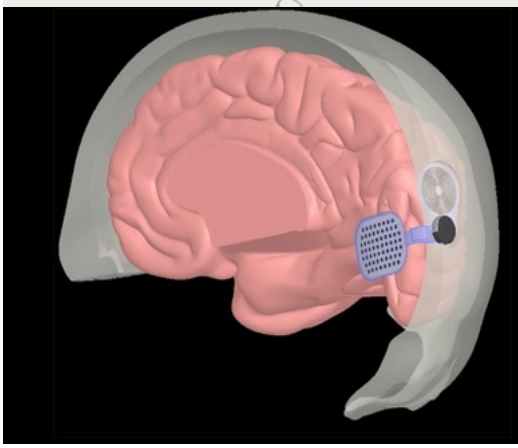
By **Andrea Park** · Feb 17, 2022 11:48am

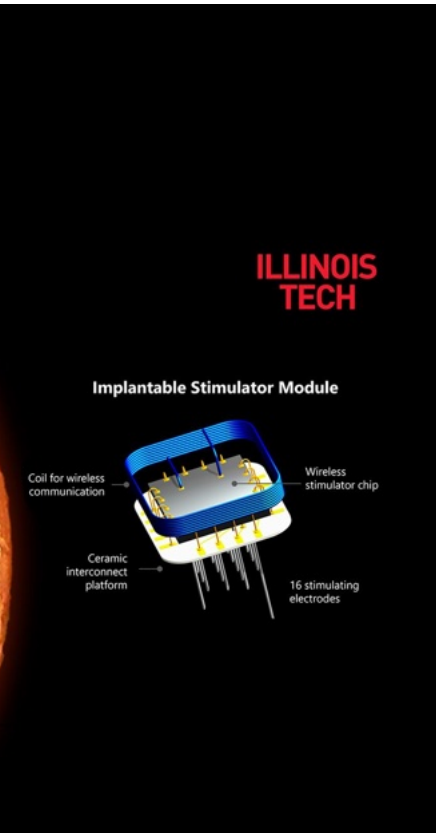
Second Sight Medial Products | implants | Retinitis Pigmentosa | investigation





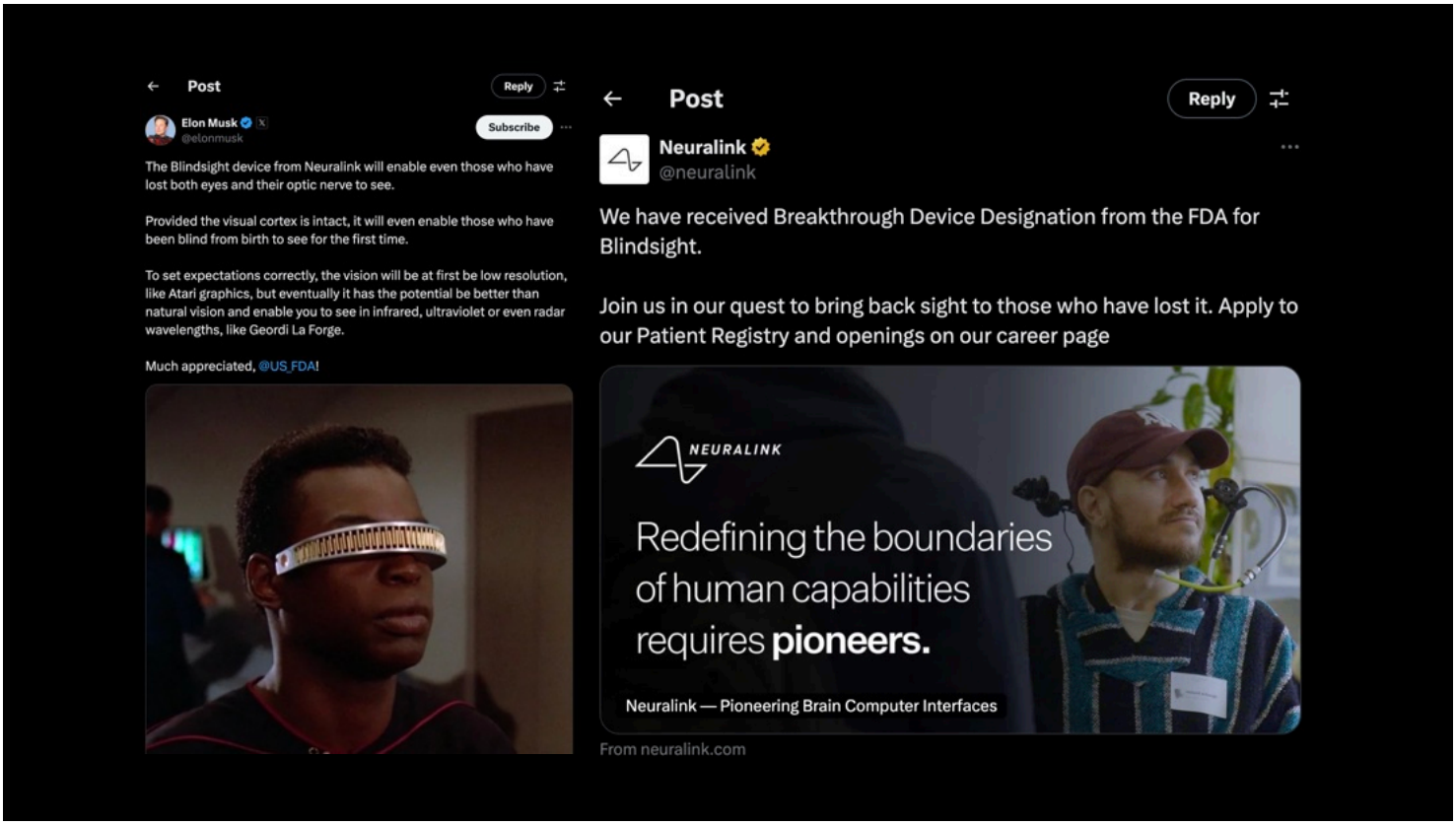


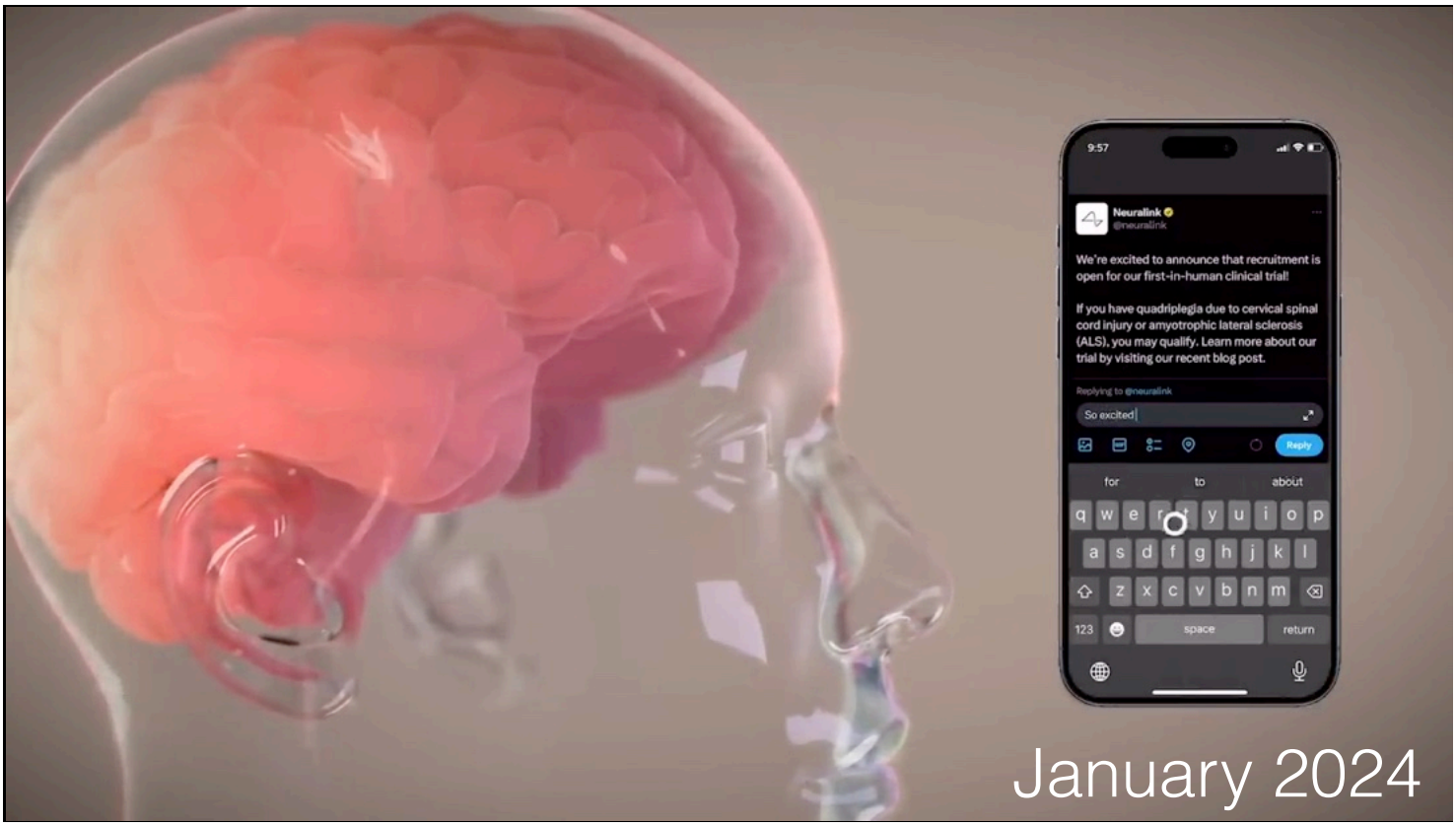


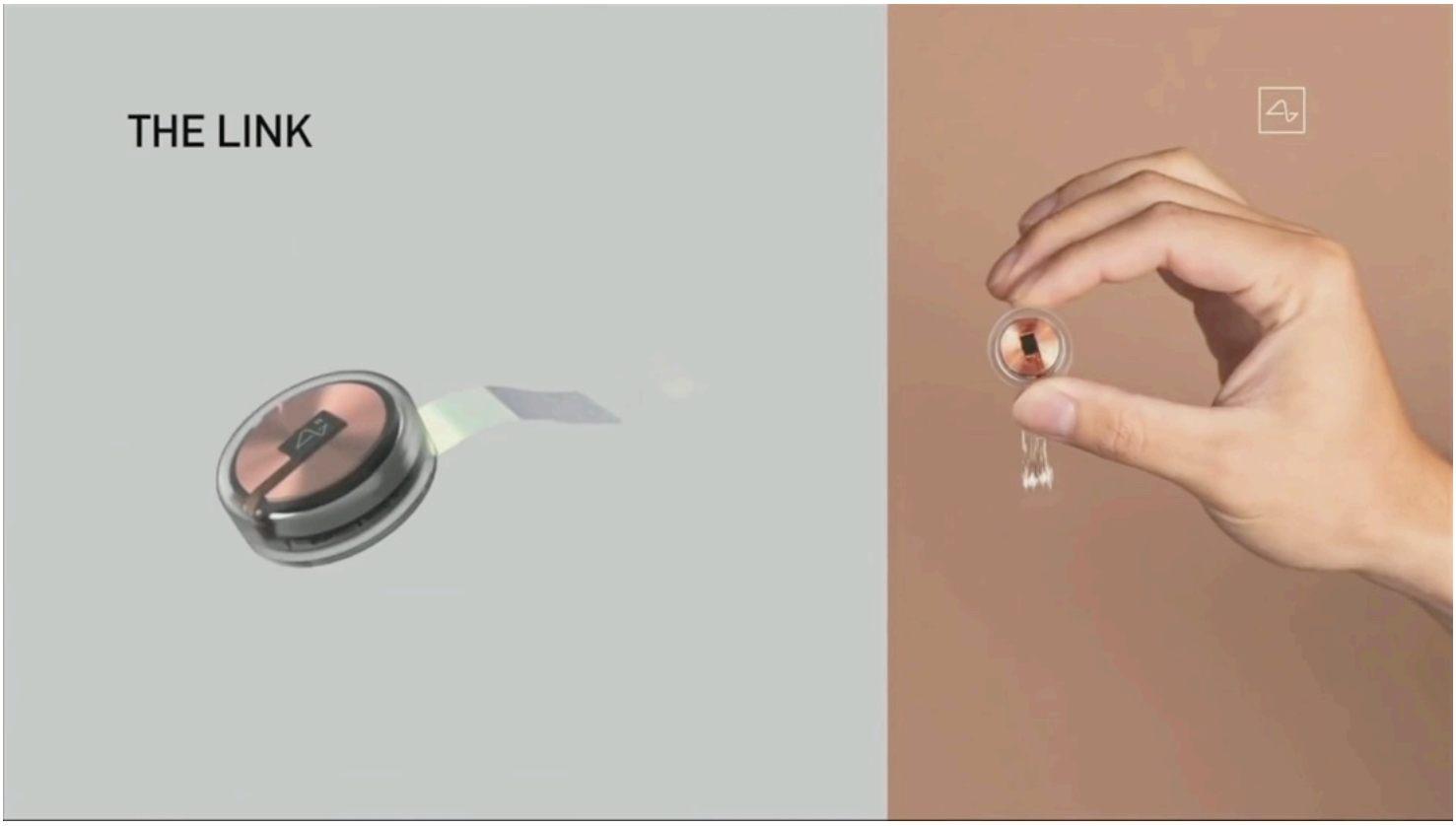


Do you know Elon Musk?

- What is Neuralink doing?




9:57
Neuralink
@neuralink
We're excited to announce that recruitment is open for our first-in-human clinical trial!
If you have quadriplegia due to cervical spinal cord injury or amyotrophic lateral sclerosis (ALS), you may qualify. Learn more about our trial by visiting our recent blog post.
Replying to @neuralink
So excited
Reply
for
to
about
space
return
January 2024


THE LINK

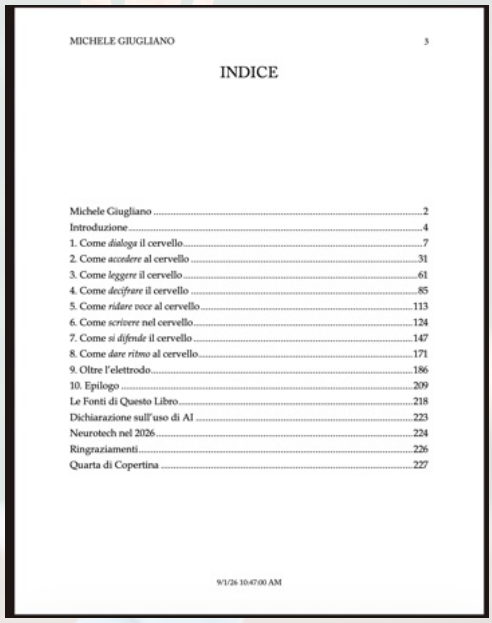
Assuming you could …“read” thoughts

- what would you (therapeutically) use it for?
- What if you can’t read but you have AI? How?

Michele GIUGLIANO

UN MICROCHIP NEL CERVELLO

Sfide aperte e confini terapeutici della Neuroingegneria



MICHELE GIUGLIANO 3

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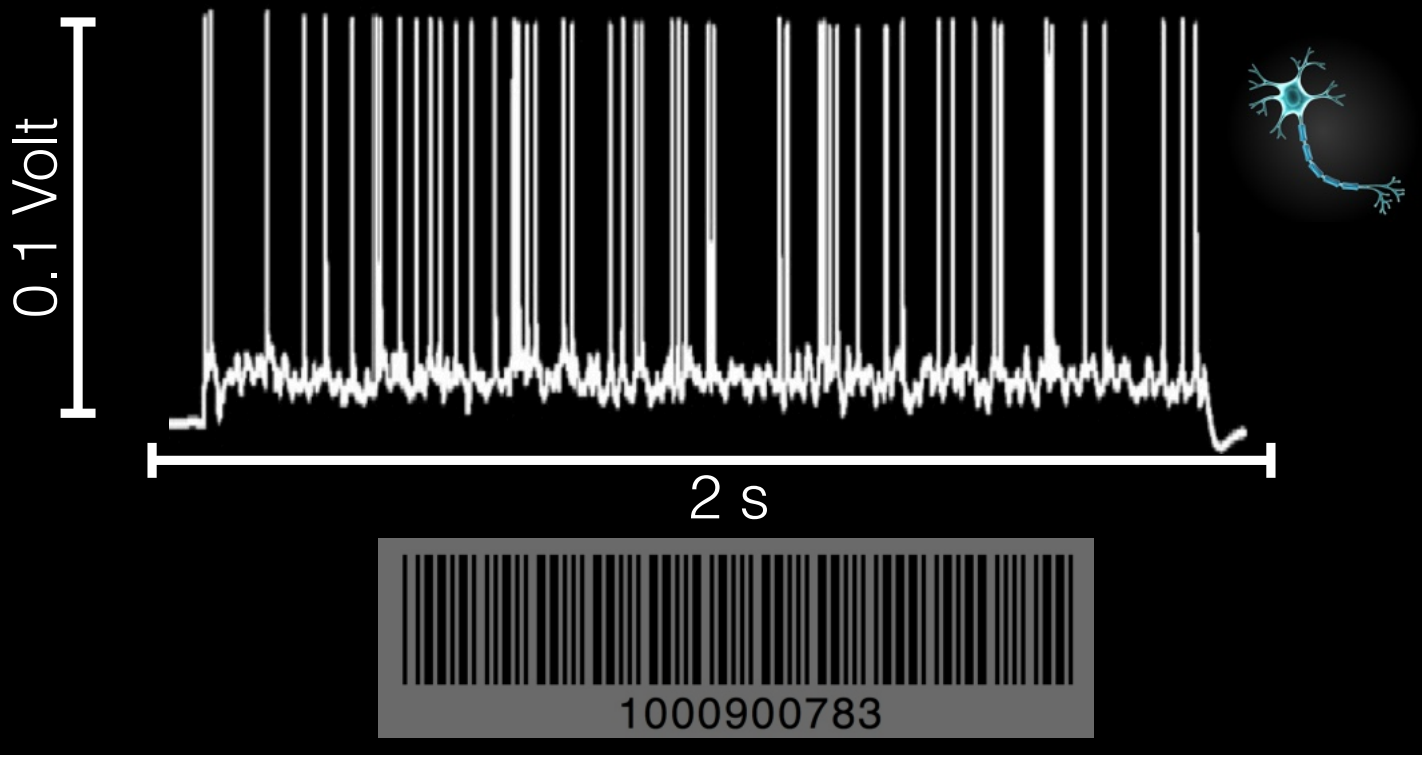
9/1/26 10:47:00 AM

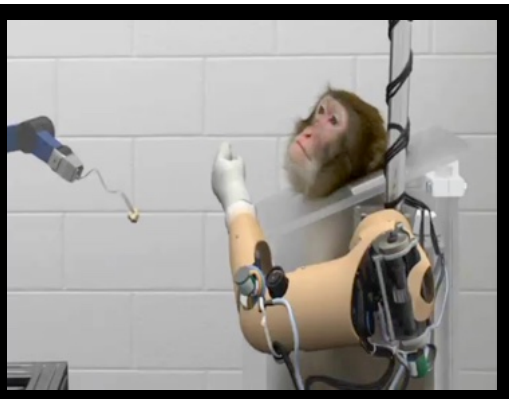
https://www.dropbox.com/scl/fi/fkcv5g1jkononevn6wh8h/microchip_full.pdf?rlkey=jtzlbsgs16z1xcks7j3r7o205&dl=0

Train of impulses from a nerve cell

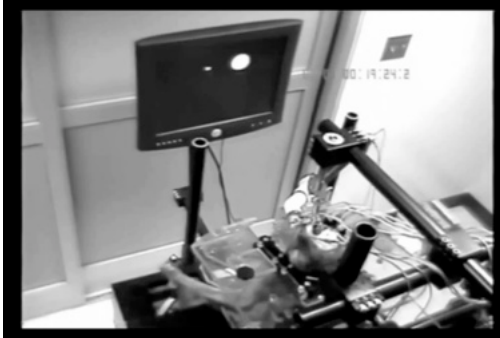


A ●–	J ●–––	S ●●●
B –●●●	K –●–	T –
C –●–●	L ●–●●	U ●●–
D –●●	M ––	V ●●●–
E ●	N –●	W ●––
F ●●–●	O –––	X –●●–
G ––●	P ●––●	Y –●––
H ●●●●	Q ––●–	Z ––●●
I ●●	R ●–●	

2003 (2008)



JOYSTICK CONTROL MODE

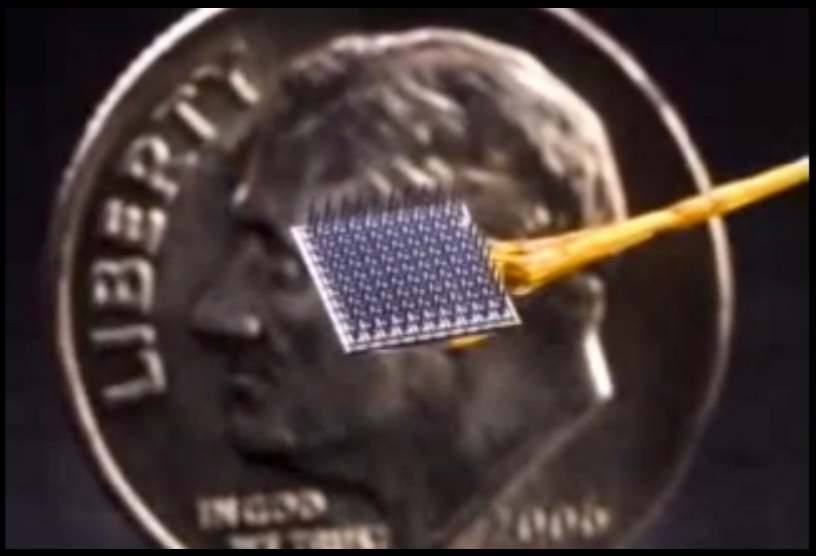


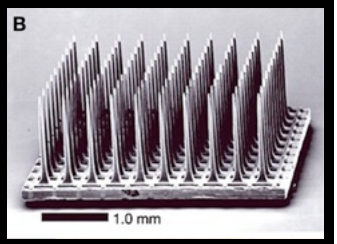
BRAIN CONTROL MODE

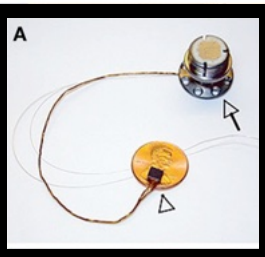
Miguel Nicolelis
Duke (US)

Electrical **Recordings**:
motor neuroprostheses (motor cortex)

2008

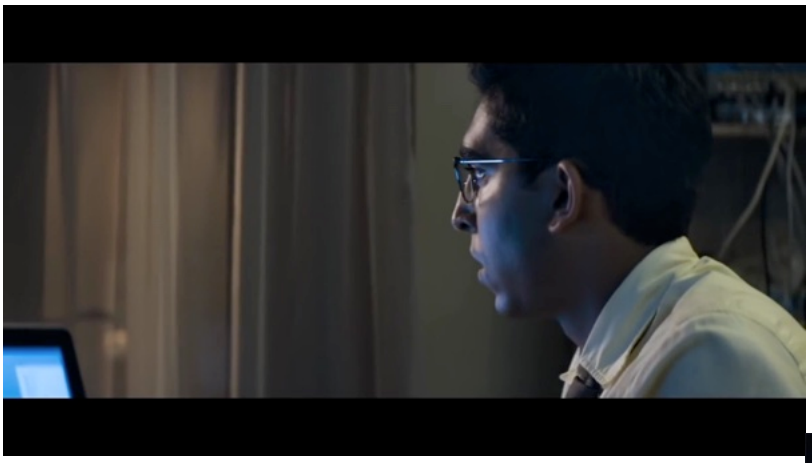






Electrical **Recordings**:
motor neuroprostheses (motor cortex)

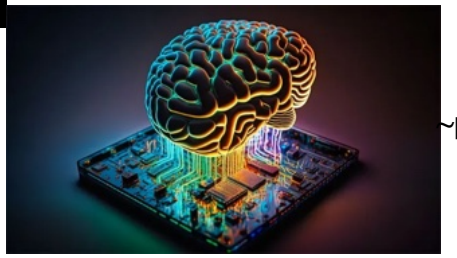
Machine Learning and Artificial Intelligence



2 gigawatt-hour (GWh) for training
~1000 households over 365 days

2 Wh/query

Neuromorphic engineering



3 Wh
~phone charger

nature communications

Article

https://doi.org/10.1038/s41467-025-64234-z

Predictive Coding Light

Received: 29 April 2025

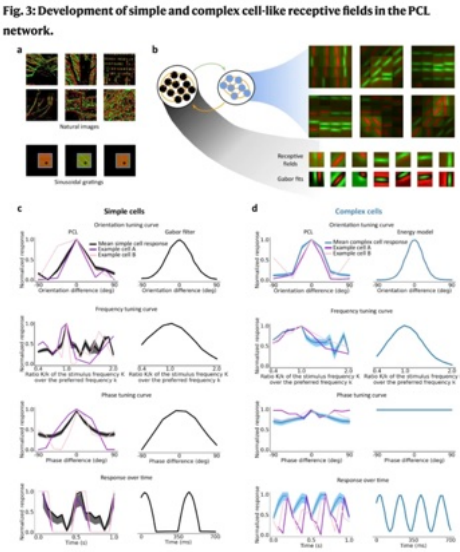
Accepted: 11 September 2025

Published online: 06 October 2025

Check for updates

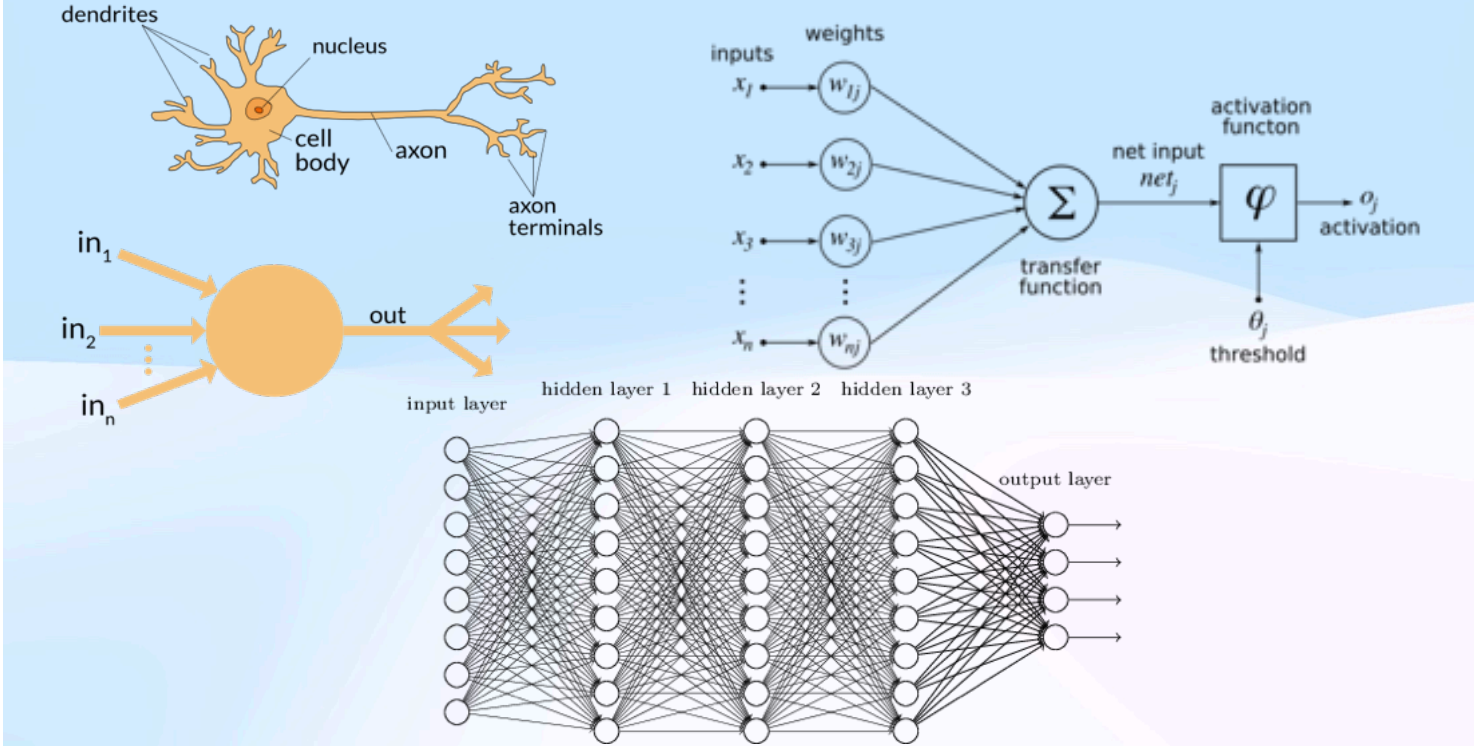
Antony W. N'dri[1], Thomas Barbier[1], Céline Teulière[1,3] & Jochen Triesch[1,2,3]

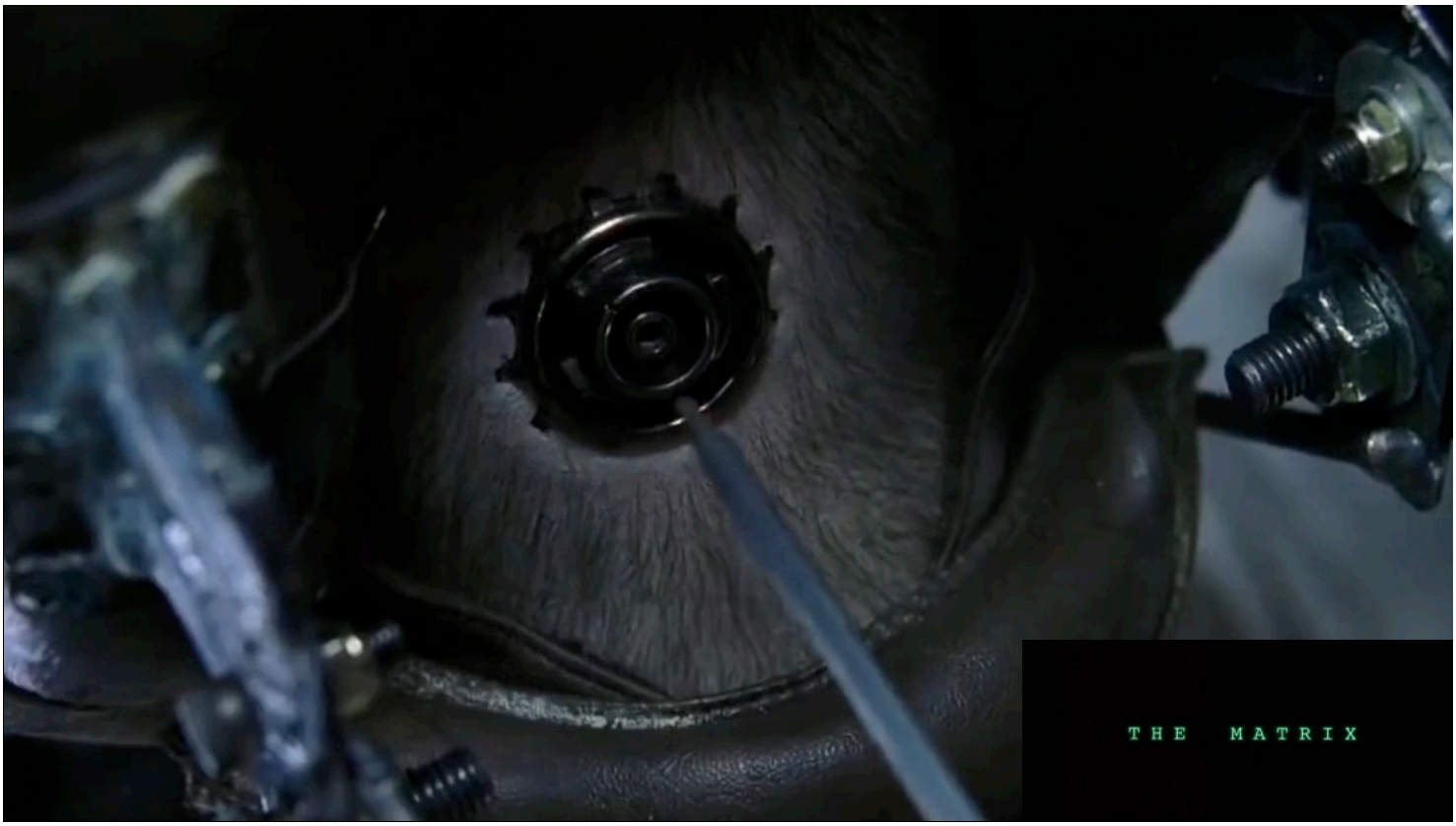
Current machine learning systems consume vastly more energy than biological brains. Neuromorphic systems aim to overcome this difference by mimicking the brain's information coding via discrete voltage spikes. However, it remains unclear how both artificial and natural networks of spiking neurons can learn energy-efficient information processing strategies. Here we propose Predictive Coding Light (PCL), a recurrent hierarchical spiking neural network for unsupervised representation learning. In contrast to previous predictive coding approaches, PCL does not transmit prediction errors to higher processing stages. Instead it suppresses the most predictable spikes and transmits a compressed representation of the input. Using only biologically plausible spike-timing based learning rules, PCL reproduces a wealth of findings on information processing in visual cortex and permits strong performance in downstream classification tasks. Overall, PCL offers a new approach to predictive coding and its implementation in natural and artificial spiking neural networks.



https://www.nature.com/articles/s41467-025-64234-z

Real neurons, Perceptron and (deep) networks





Biohybrid computation - “wetware”

BBC

Home News Sport Business Innovation Culture Arts Travel Earth Audio Video Live

Scientists grow mini human brains to power computers

3 days ago

Share Save

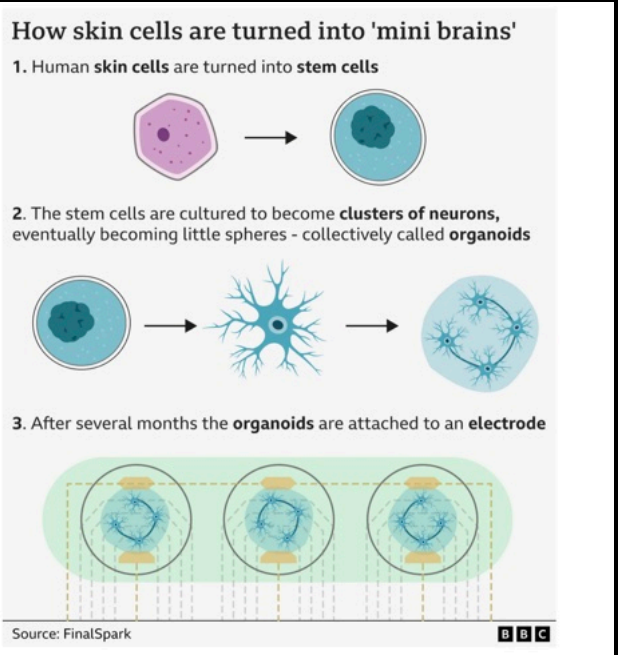
Zoe Kleinman Technology editor

The building blocks for a biocomputer growing in a lab

It may have its roots in science fiction, but a small number of researchers are making real progress trying to create computers out of living cells.

Welcome to the weird world of biocomputing.

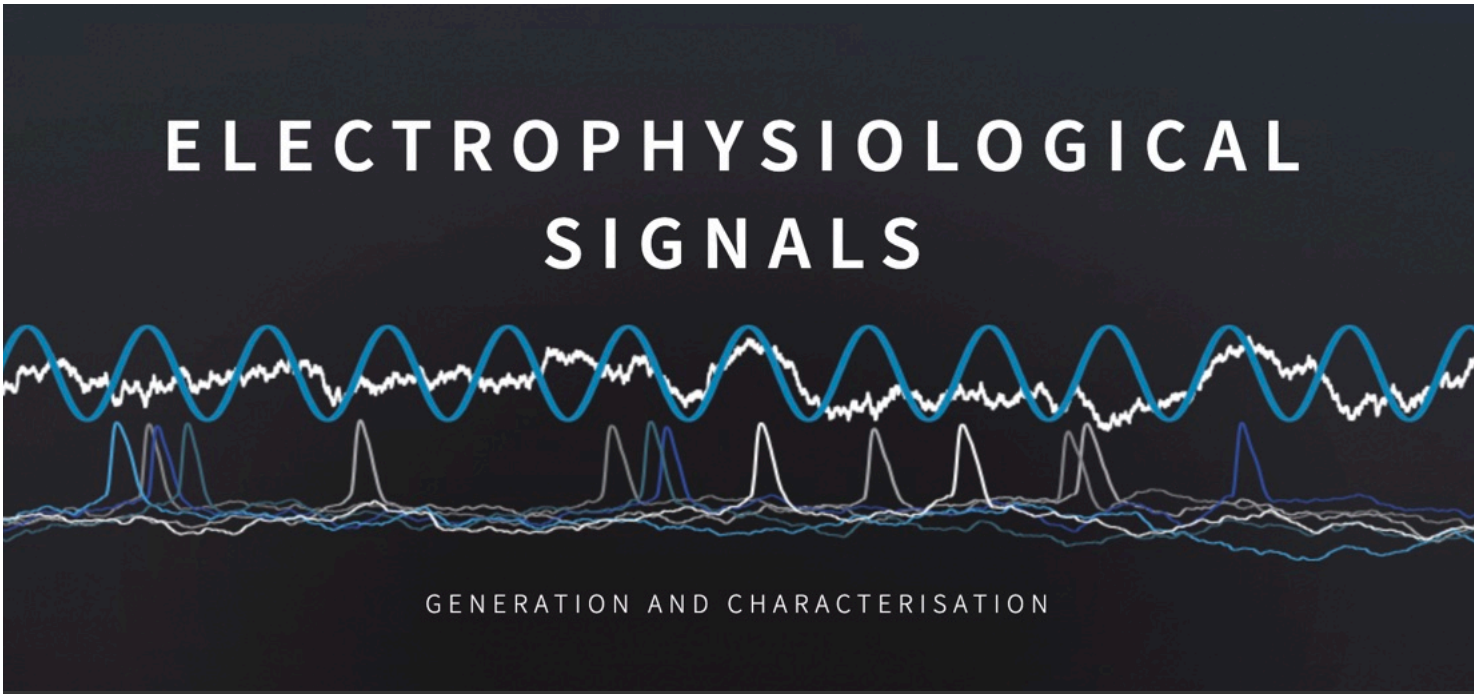
Among those leading the way are a group of scientists in Switzerland, who I went to meet.



https://finalspark.com/live/

Biohybrid computation - “wetware”

…in 2001 !



Prof. Ing. Michele GIU**G**LIANO, PhD

Introductory Class - continues

ATTENDANCE TRACKING: **today's code is DC8R9**

(for my own statistical purposes)

Download for iPhone

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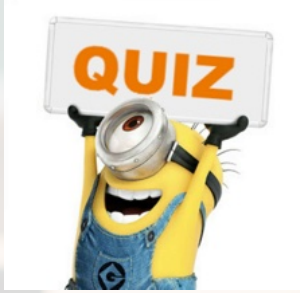
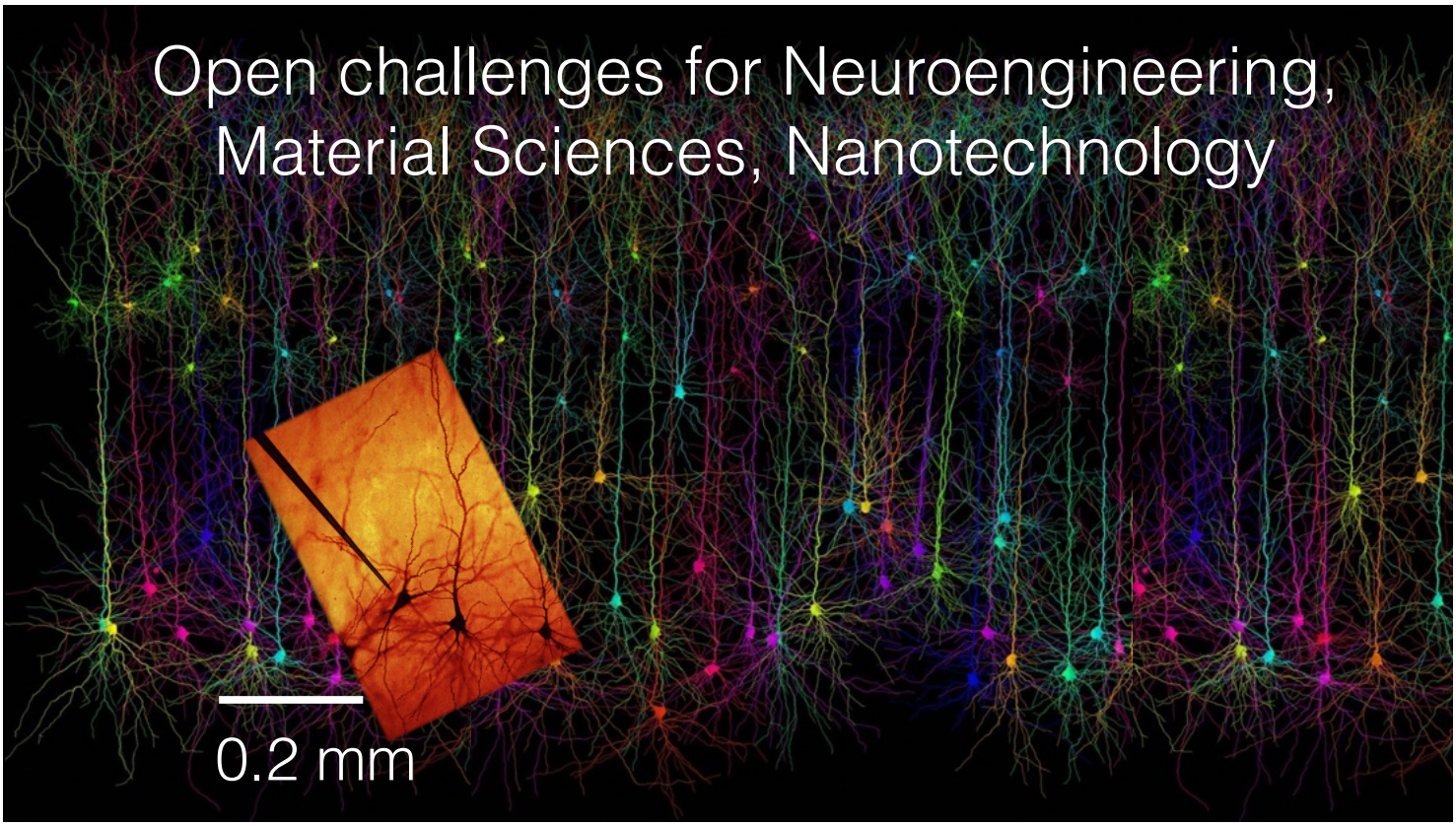
https://www.unimore.it/it/servizi/unimore-app

How big is big?

- What is roughly the size of a neuron (body)?
- What is the size of a DBS electrode?

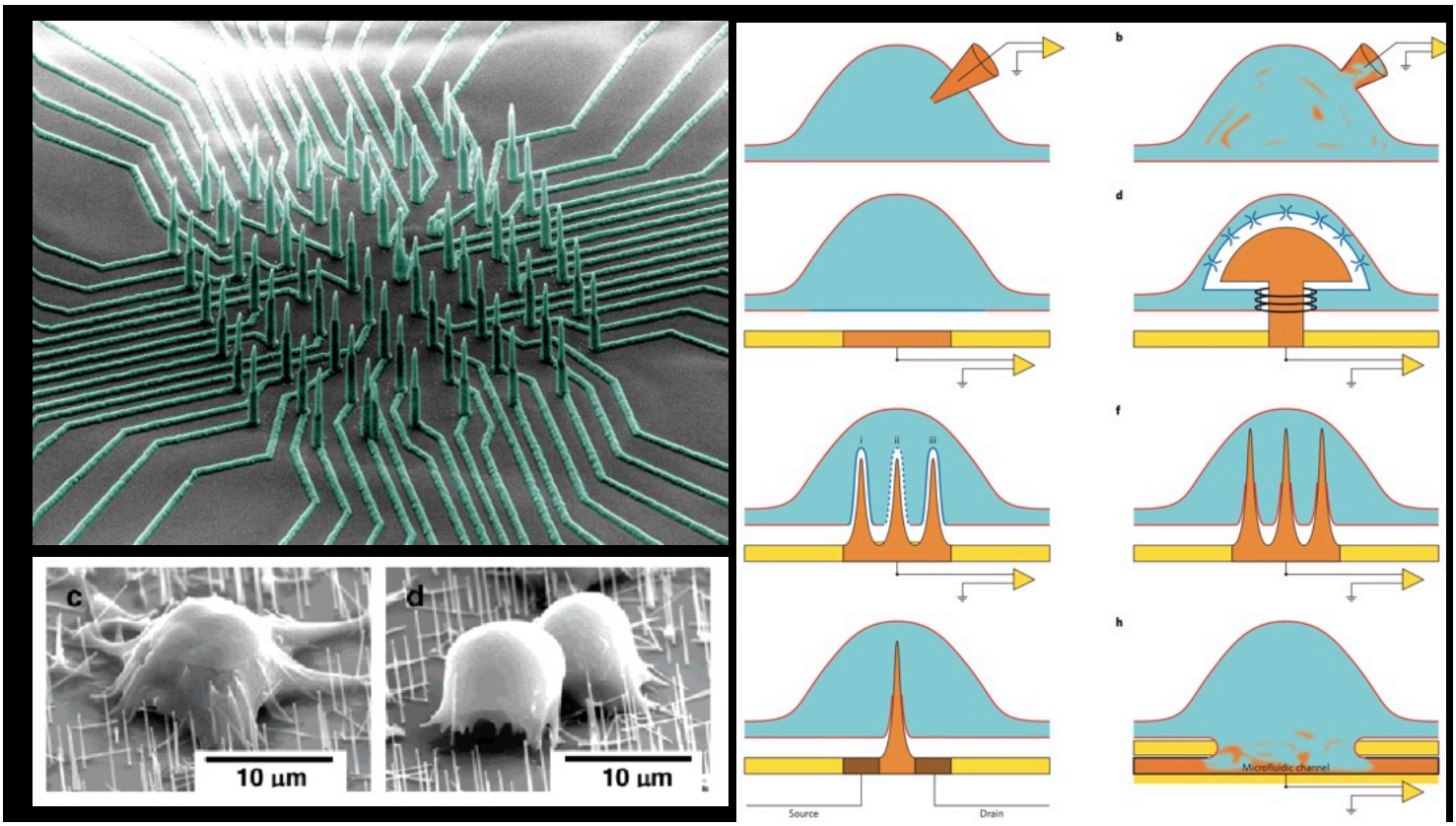


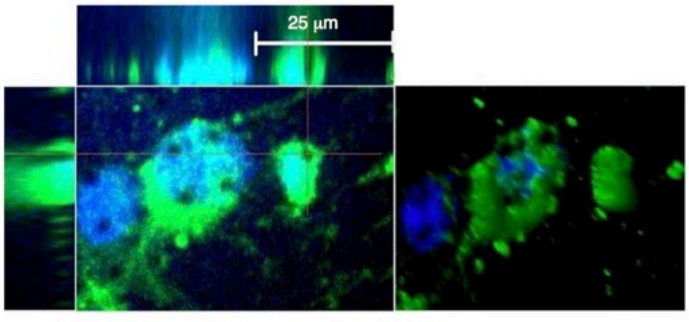


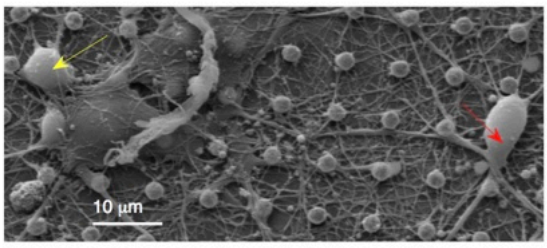
Challenges of Neuroengineering

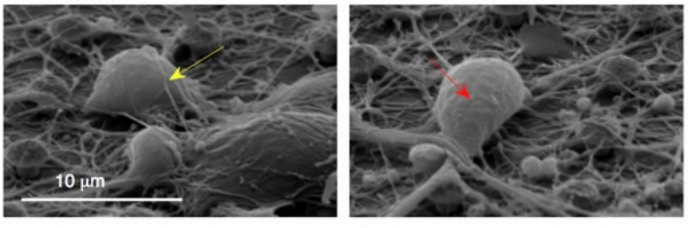
- **NON selective** stimulation
- Discontinuous and **NON selective** recordings
- Foreign-body tissue rejection reaction, inflammation

- **decoding** information
- design and simulate **mathematical models** of nerve cells, circuits, systems





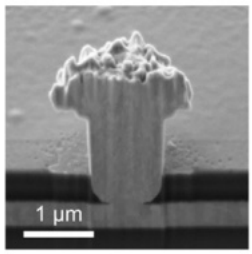


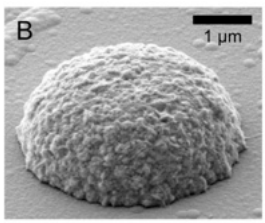


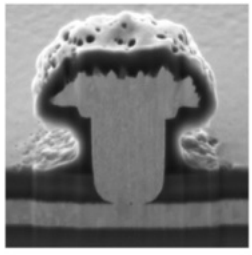
Peter JONES

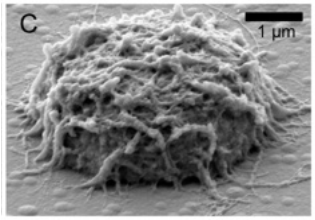
Anastasiya MOSKALYUK

Micha SPIRA

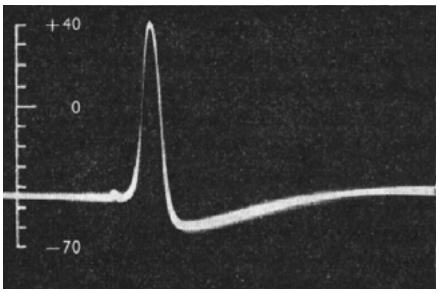


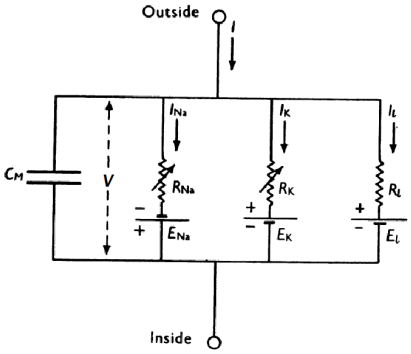




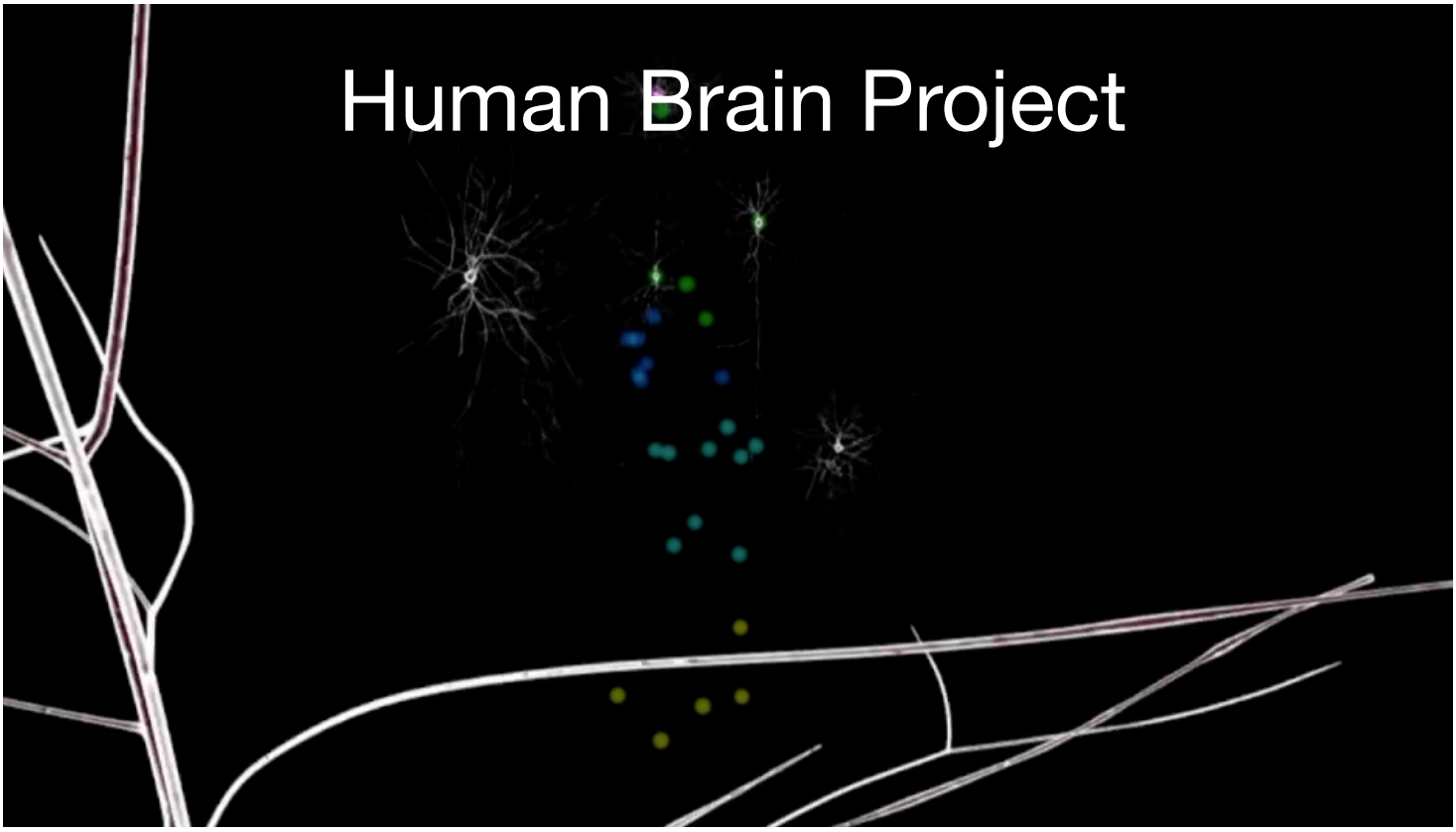


Jones et al. 2020



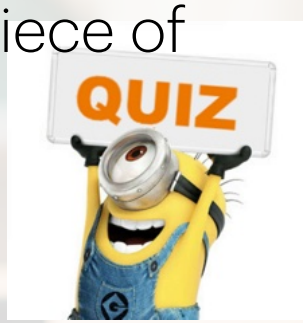


$$\frac{dV}{dt} = -\frac{1}{C_M}\{\bar{g}_K n^4 (V - V_K) + \bar{g}_{Na} m^3 h (V - V_{Na}) + \bar{g}_l (V - V_l)\}$$



High Performance Computing

- What is a super-computer?
- How would you run a “simulation” of a piece of brain on a super-computer?



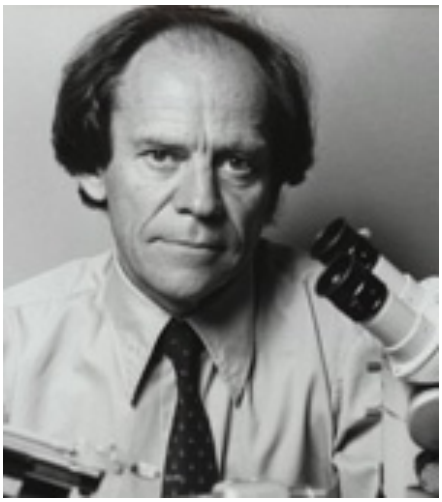
Examples

of (electrophysiological) brain signals

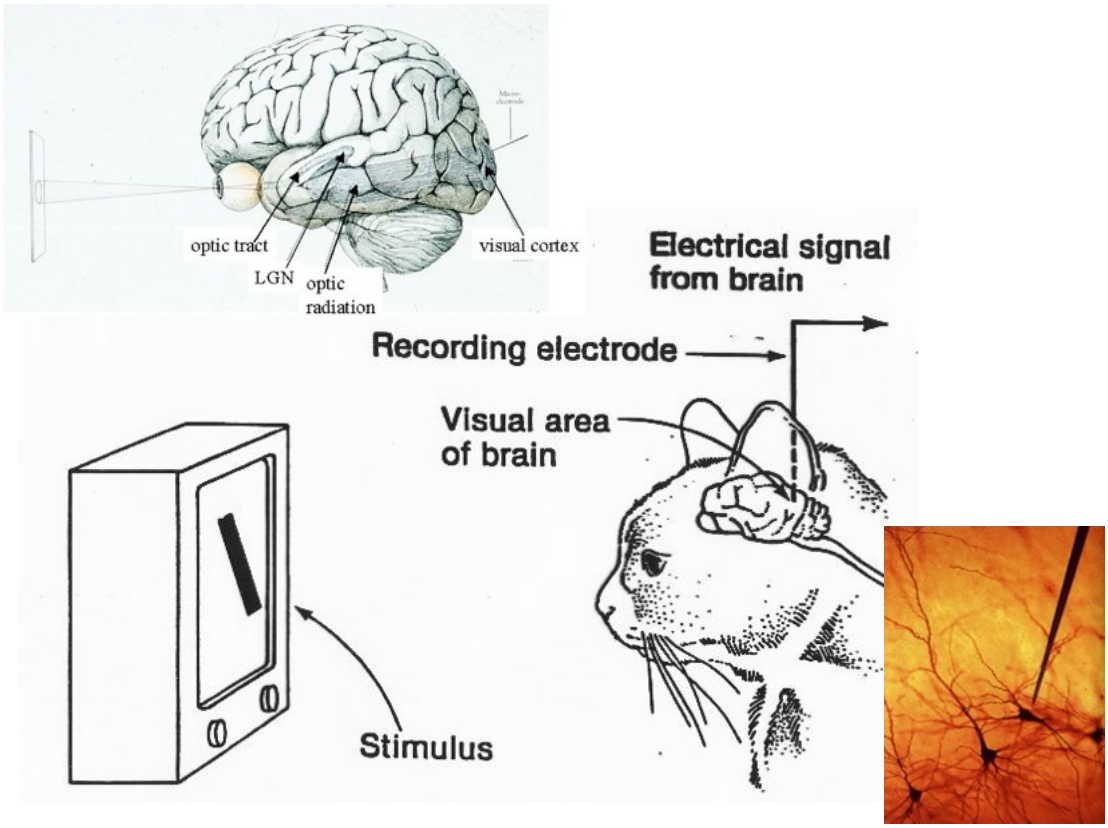
Mammalian Visual System

David Hubel

Torsten Wiesel



Nobel prize for Physiology or Medicine (1981)



Ganglion cell - Pay attention

- when/what is this neuron “responding” to?
- what are the features of the stimulus that… excite that neuron?
- what “computation” performed on the image?

Ganglion cells / LGN cells

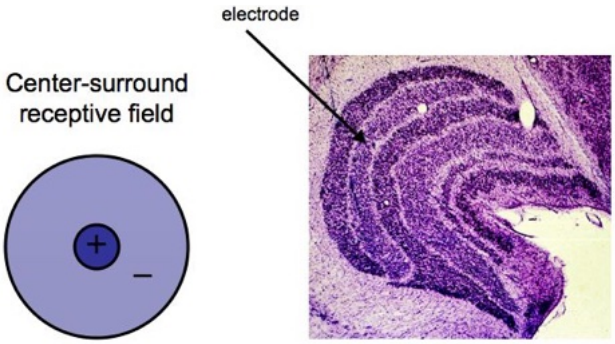
center ON - surround OFF

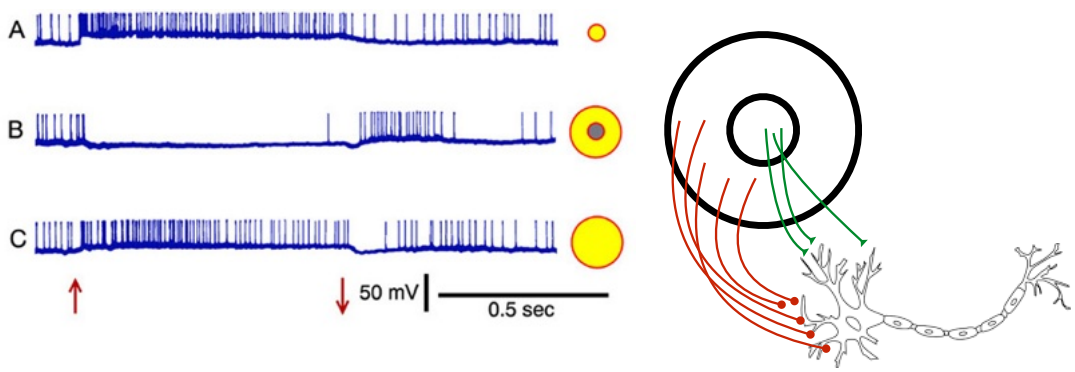
- exist a specific, circular receptive field
- no orientation selectivity
- cells fires action potentials spontaneously, at low rates
- cell fires action potentials when the light is in the center
- cells suppresses its firing when the light is the surround
- cells fires action potentials if light is everywhere
- firing responses are not sustained in time, fatigue, adaptation
- switching off the light over the surround causes rebound firing
- other cells show center OFF - surround ON

It is a contrast feature detector! (detects and relays info on **edges**)

Ganglion cell - Circuit?

- How could you build a similar circuit?



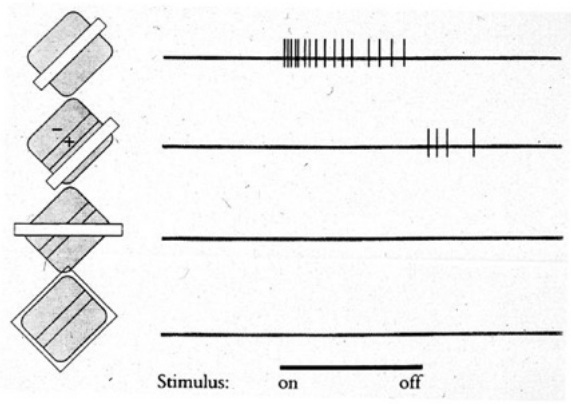


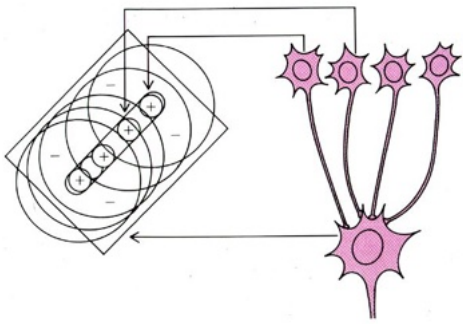
Cortical cell - Pay attention

- when/what is this neuron “responding” to?
- what are the features of the stimulus that… excite that neuron?
- what “computation”?

V1 (simple) cells

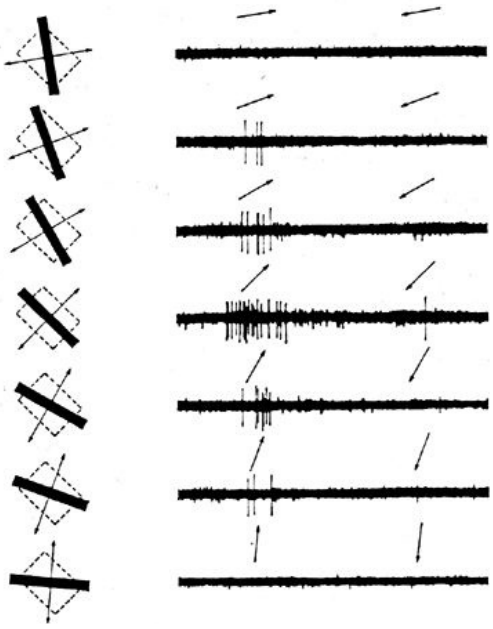
orientation selective, center-surround



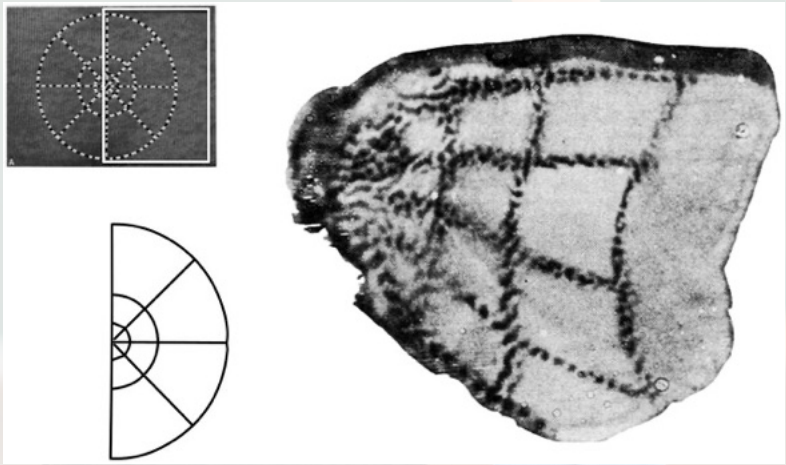


V1 cells

orientation and direction selective



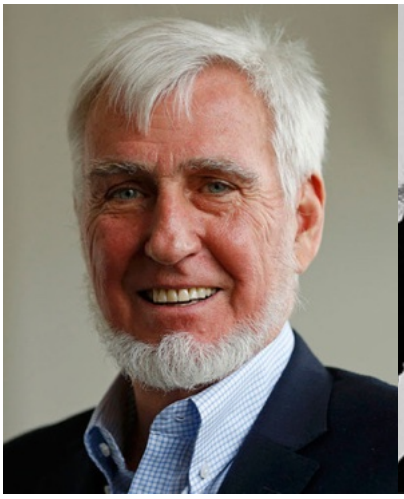
What is Retinotopy? (Tonotopy)



Fovea is over *represented!*

Hippocampus, Enthorinal Cortex: cognition/memory

O'Keefe

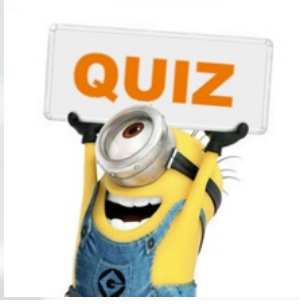


Moser & Moser

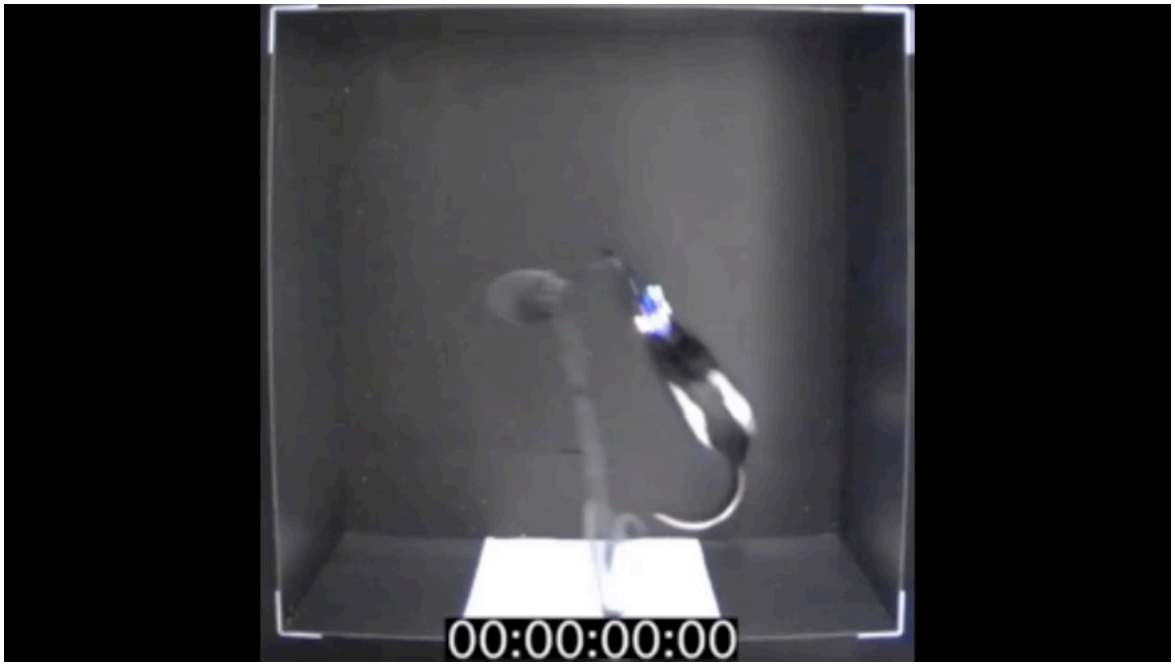
Nobel prize for Medicine (2014)

Hippocampal cell - Pay attention

- when/what is this neuron "responding" to?
- what "computation" is it useful for?



Hippocampal “place” cells



Hippocampal “replay” of memories?
Yes! During stillness and (non)-REM sleep!

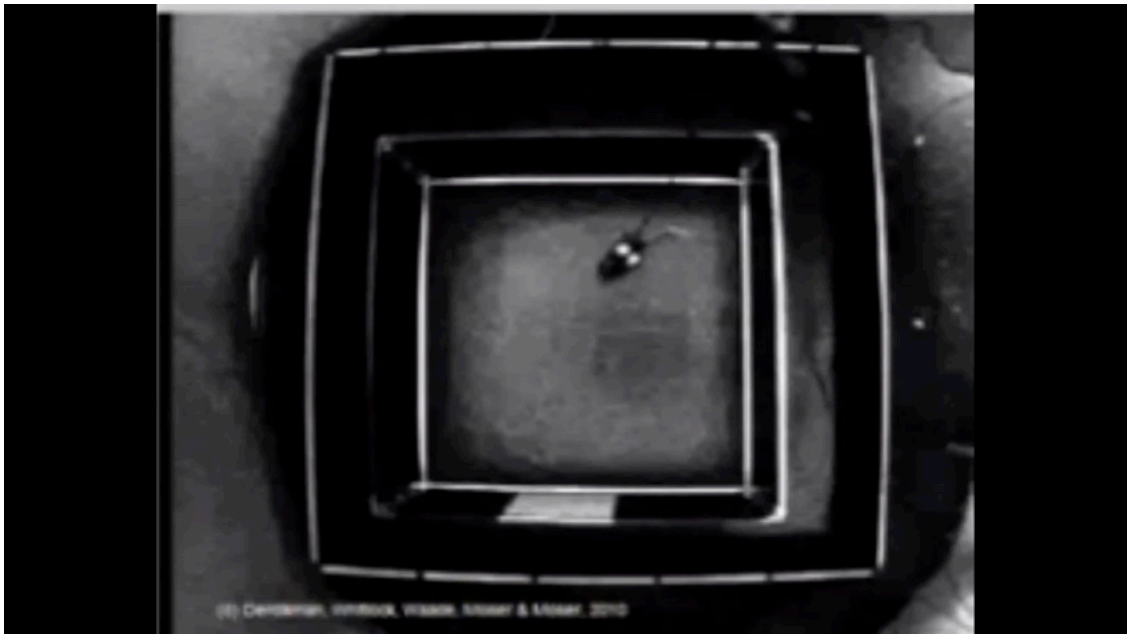
Reading the Minds of Rats I Matt Wilson I TEDxCoconutGrove
(20 mins TED Talk)

https://www.youtube.com/watch?v=Vf_m65MLdLI

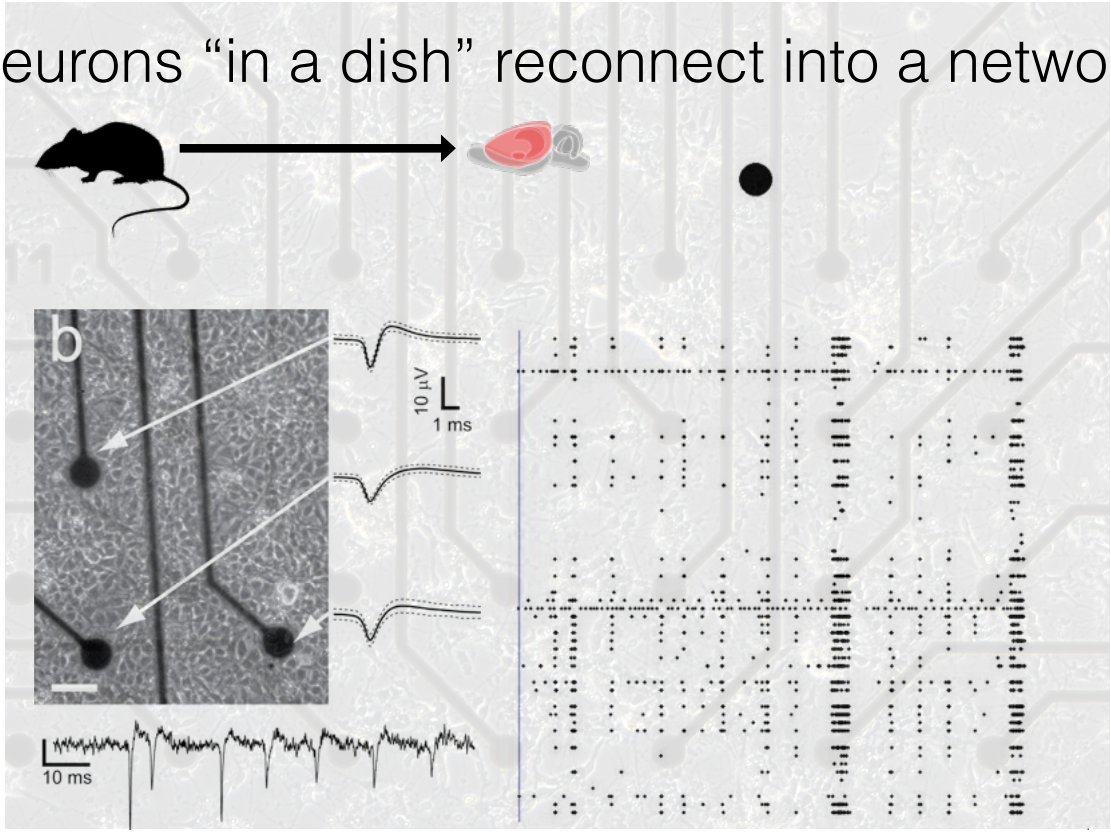
Enthorinal cortex - Pay attention

- when/what is this neuron “responding” to?
- what “computation” is it useful for?

Enthorinal cortex

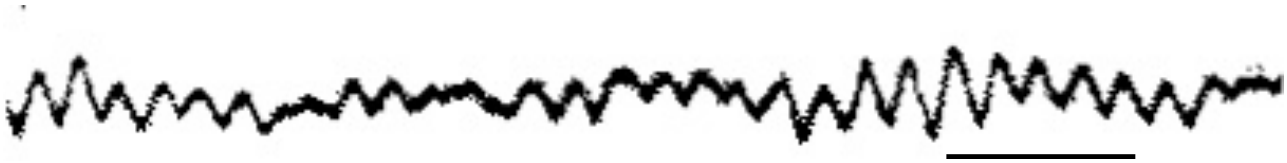


Neurons “in a dish” reconnect into a network



Hans Berger
(1873-1941)

Electroencephalography
EEG



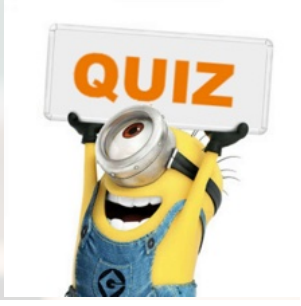
1924

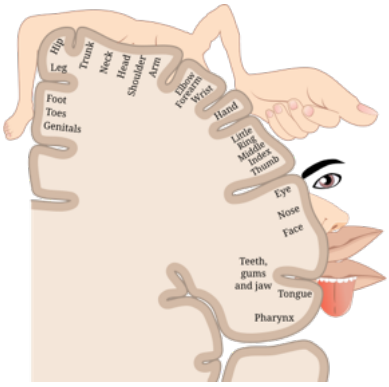
EEG Rhythms

- EEG Rhythms e.g.
  - Categorization of rhythms based on frequency
    - Beta: Greater than 14 Hz, activated cortex
    - Alpha: 8-13 Hz, quiet, waking state
    - Theta: 4-7 Hz, some sleep states
    - Delta: Less than 4 Hz, deep sleep
  - Deep Sleep
    - High synchrony, high EEG amplitude

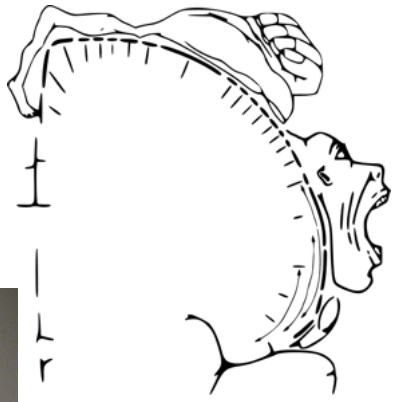
Physiological/Pathological rhythms

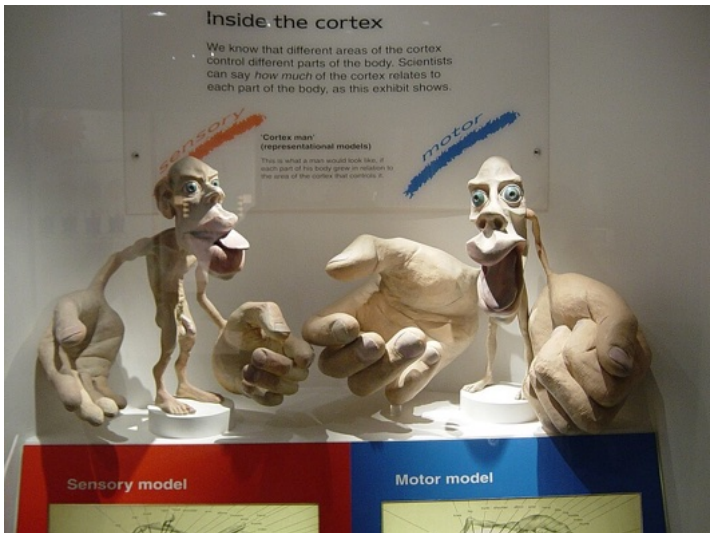
- What is Penfield's cortical *homunculus*?



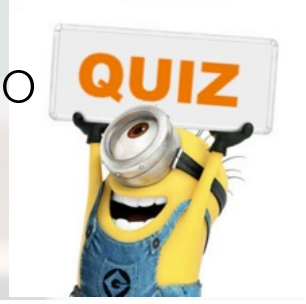


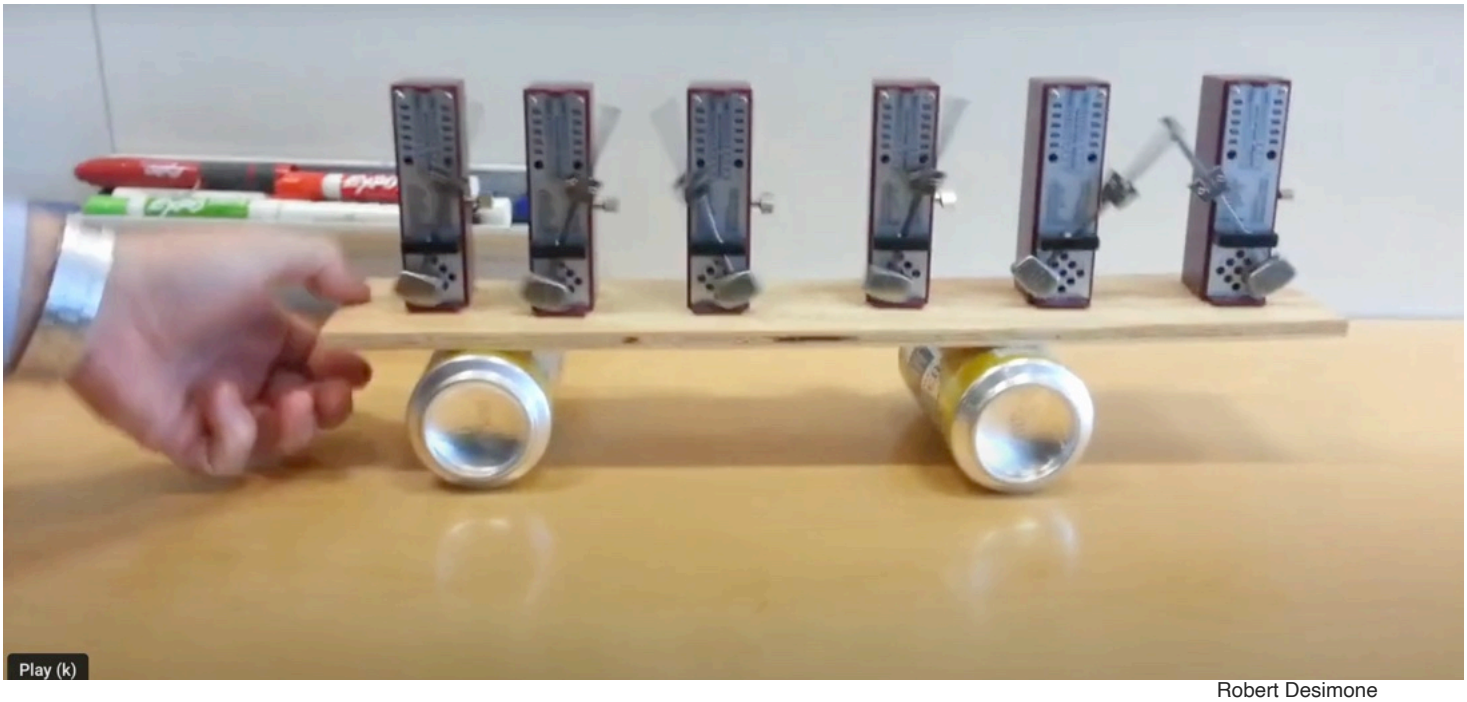
Homunculus





- Why “hands” and “lips" are *deformed*?
- What would you experience if a wave of abnormal (synchronous) electrical activity invaded the entire cortex?
- How could it propagate from one spot to its neighbors?





Robert Desimone

EEG signals relevance?

Abnormal rhythms & synchronization

epilepsy (~1% of the population)

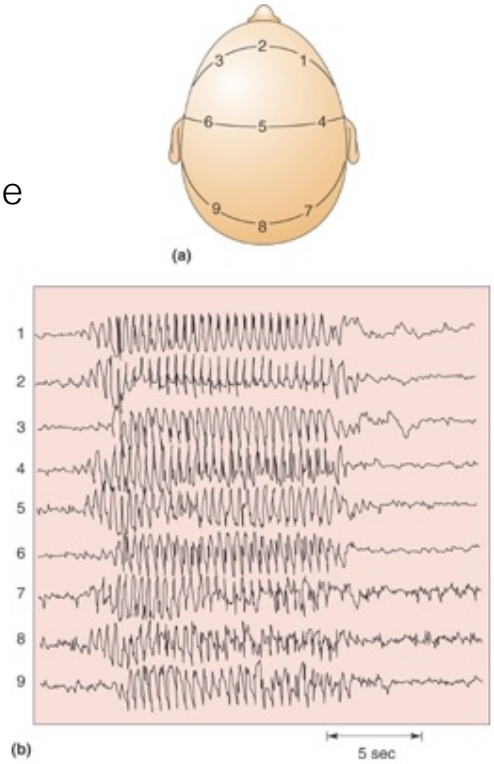
It is a symptom, not a single disease (60% is treated by drugs): spontaneous and recurrent occurrence of **seizures**, a sudden excessive discharge of CNS electrical activity, causing unexpected changes in behavior, motor function, sensation, consciousness.

400 B.C. Hippocrates; Gospel; Witch hunting (~1400 A.D.)

Epilepsy

- seizures lasting <10 min do not cause brain damage
- no correlation to IQ, to violence..
- not necessarily inherited - unknown causes

- *petit* mal (absence seizures)
- *grand* mal (tonic-clonic seizures)



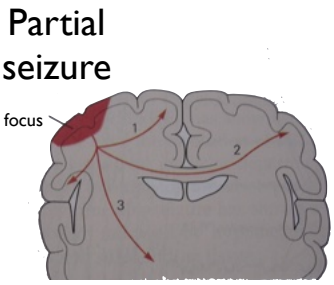
Seizure classifications

- Focal s. (known also as "partial", spatially constrained)
  restricted to only one part of the cortex
- Generalized s. (involving thalamo-cortical circuits??)
  spread to the entire cortex, simultaneous

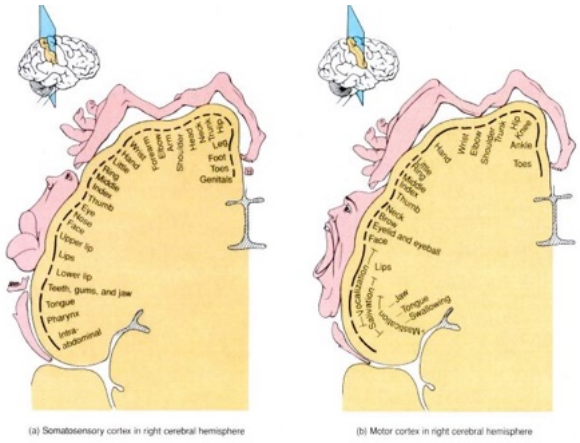
- Simple s.
  consciousness is unaffected (full awareness, although possible impairments in speech)
- Complex s.
  consciousness is impaired during the event (memory system, emotion processing,..)

- Motor s.
  motor areas are involved - tonic - clonic - atonic
- Sensory s.
  sensory areas are involved

(focal, motor) Seizures



Here, not spreading to other areas.



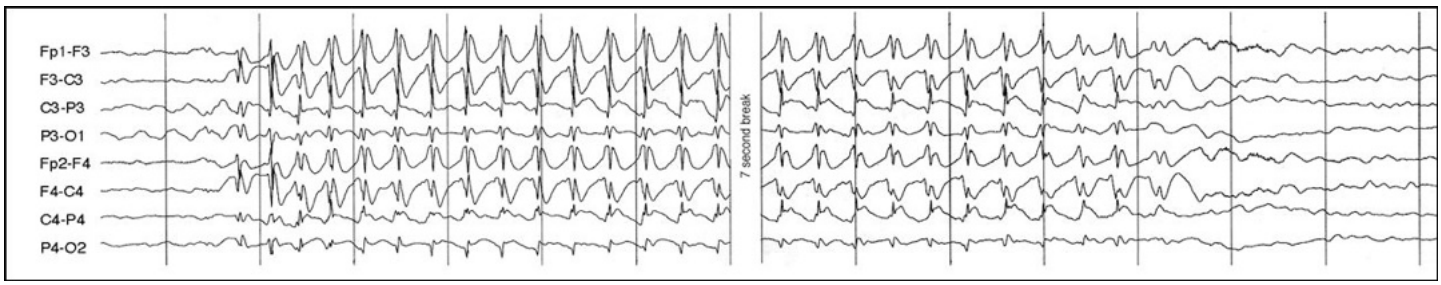
Cortical *homunculus* (Penfield's)

intermittent (clonic) motor activity, consciousness unaffected, unilateral character, localization

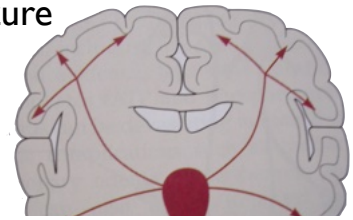
Any idea why it is intermittent ???

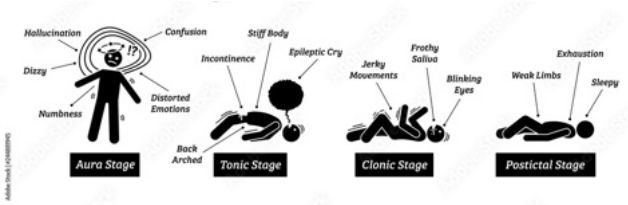
(generalized) *Absence* Seizures

- 2 - 30 seconds episodes, absence, twitching eye movements
- 1-2 times - 100 times a day
- prepubertal age
- stereotypical spike & wave, 3 Hz

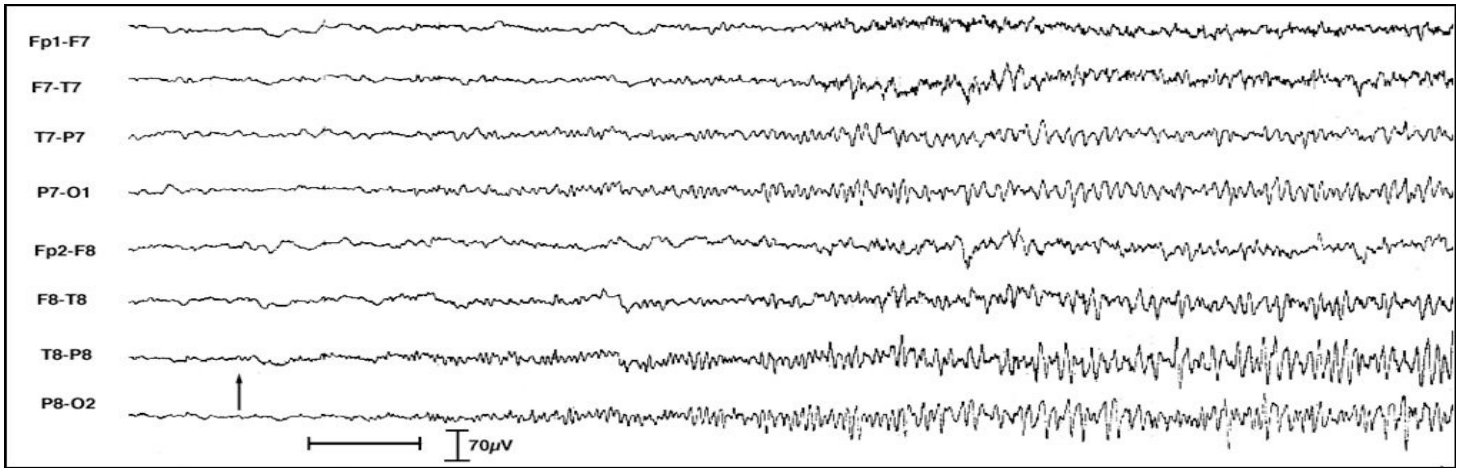


Generalized seizure

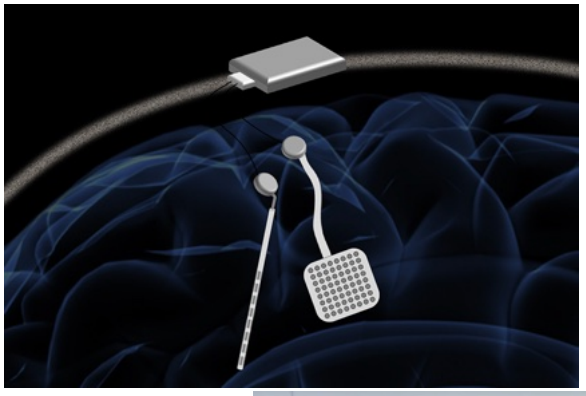


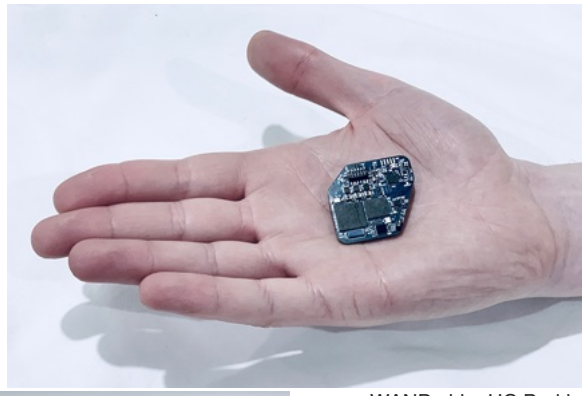


Generalized seizure

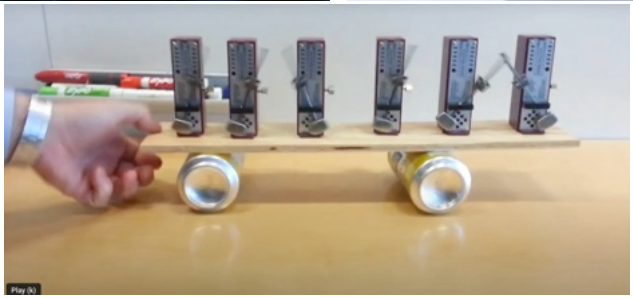


Brain pacemakers… for epilepsy

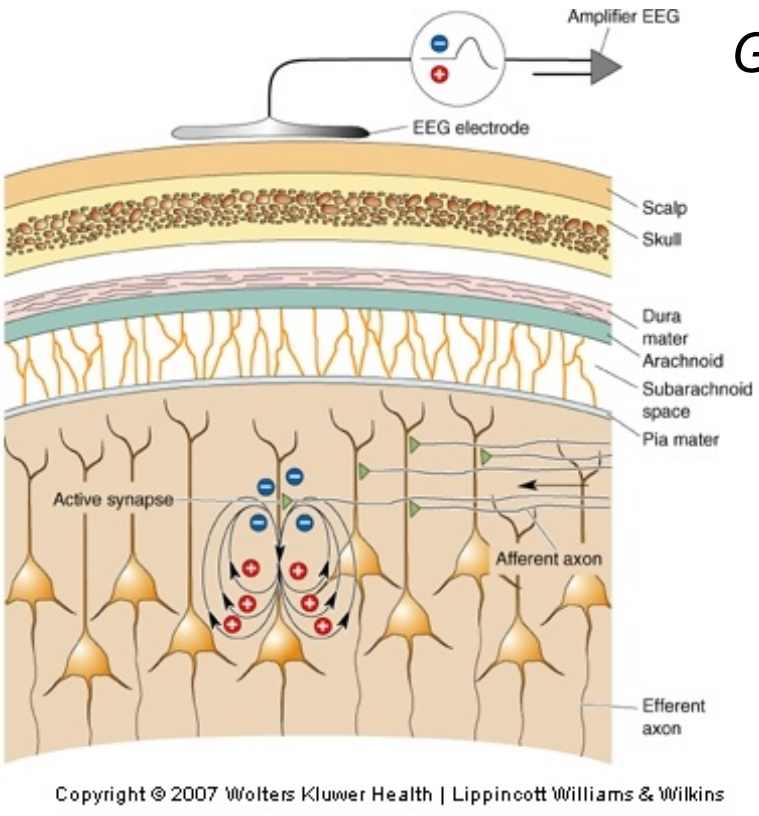




WAND chip, UC Berkley



Robert Desimone



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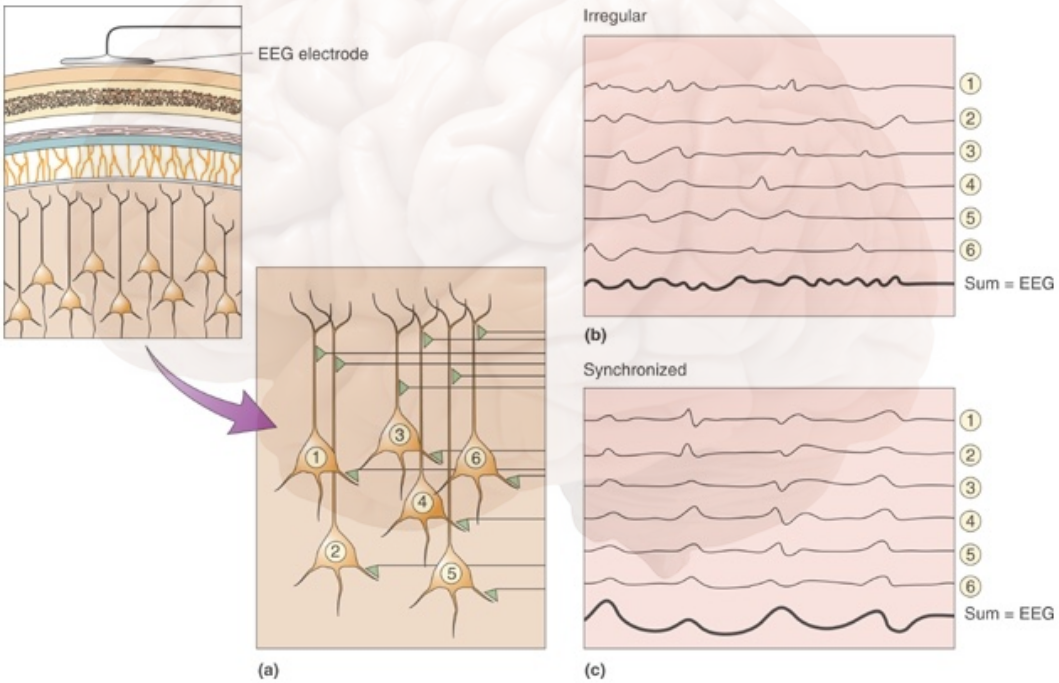
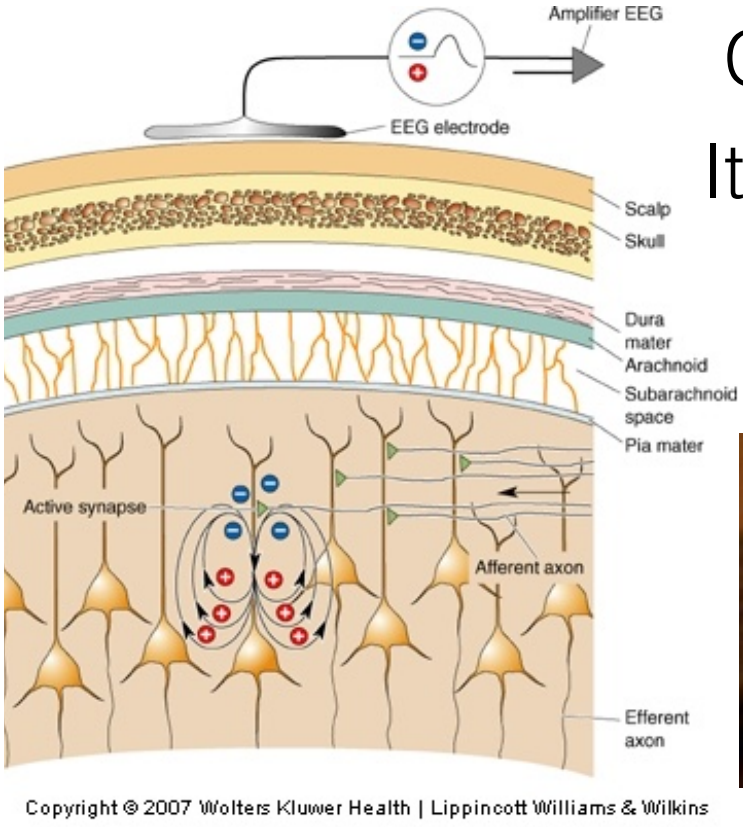
Generation of EEG?

…by the synaptic inputs to the pyramidal neurons (~80% of cortical cells)

Rhythms in EEG: how?





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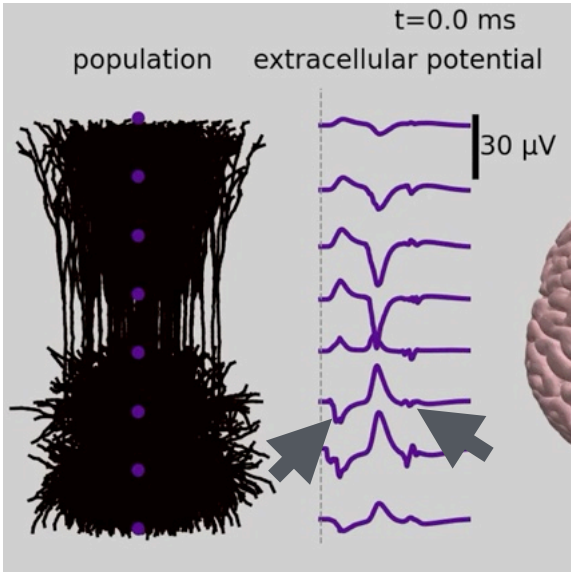
Complex generation!

It calls for an **in-depth** understanding!

Intracellular signals, extracellular signals, geometry, large population, dynamical state…



Whatever those cells do, the extracellular potentials are … difficult to understand.

Torbjørn V. Ness's YouTube Channel